Domain Adaptation Support Tensor Machine: An Extended STM for Object Recognition Using Cross-Source Heterogeneous Remote Sensing Data

Tong Gao, Lingjia Gu, *Member, IEEE*, Hao Chen, *Member, IEEE*, and Bin Zhou, *Senior Member, IEEE*

***Abstract*— Multisource remote sensing data observed from sensors with different resolutions and physical properties will present heterogeneous tensor structures and diverse feature distributions, thus posing a significant challenge for building an effective classifier for cross-source object recognition. The representative support tensor machine classifier can inherently preserve tensor structure information of remote sensing data and obtain effective recognition ability, while it can only handle same-source and same-distributed homogeneous data and fail to deal with cross-source heterogeneous remote sensing data with complex structures and various distributions. Therefore, the domain adaptation support tensor machine (DA-STM) is proposed to learn a uniform model for cross-source object recognition. To process heterogeneous tensor from different sources, multiple factor matrices with different modes are constructed to eliminate the structural differences and reduce distribution discrepancies for multisource heterogeneous data. To excavate shared classification information across sources, the shared core tensor is established to learn the classification hyperplane jointly using multisource data, and the adaptive sample labels are then embedded into the model to recover the class information during model training. To ensure efficient training, the decomposition algorithm is developed to accelerate the solving of dual problem of DA-STM. In addition, to improve classification performance as the acquirement of sequential samples, the proposed DA-STM is further upgraded to an online version to update the classification parameters dynamically. Using multi-resolution and multi-angle optical images as well as multi-angle SAR images, experimental results demonstrate that the proposed DA-STM can obtain better recognition results than typical domain adaptation methods.**

***Index Terms*— Cross-source object recognition, decomposition algorithm, domain adaption, domain adaptation support tensor machine (DA-STM), online DA-STM (ODA-STM).**

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I. Introduction

THE development of satellite sensors allows people to acquire multisource remote sensing data with different resolutions, imaging angles, and physical properties, thus supporting more effective object recognition applications [1], [2]. At the same time, it is also worth mentioning that multisource data obtained from different sensors present heterogeneous data structures (i.e., different spatial and spectral resolutions) and different feature distributions. Due to the basic assumption of the classifier methods, i.e., the training data and test data must present the same data structure and meet the independent identically distribution (I.I.D.) requirement [3], the classical classifier trained by data from a specific source can hardly be applied in data from the other sources. Therefore, it is necessary to develop the classifier technology to adapt effectively to different sources of remote sensing data with heterogeneous structures and various distributions in order to support cross-source object recognition.

Facing these requirements, researchers developed domain adaptation techniques to improve the adaptability of classifiers for cross-source recognition, including cross-source deep learning methods and cross-source machine learning methods. Deep learning methods have been widely applied to different remote sensing processing tasks recently, including sample augmentation [4], hyperspectral image (HSI) classification [5], change detection [6], and object detection [7]. For cross-source recognition cases, the deep learning methods always build specialized cross-domain loss modules, e.g., distribution alignment module [8] and adversarial learning module [9], to reduce the discrepancy between data from different sources and achieve cross-source object recognition. For example, domain adversarial neural networks (DANNs) utilize the gradient reversal layer to adjust the distribution between different sources so that the networks can recognize samples from arbitrary sources [10]. The minimax entropy (MME) networks exploit the feature prototypes to align the features from different sources and increase their discriminative power [11]. By integrating different types of classification losses, task-specific domain adaptation networks are developed to learn adversarially domain invariant features across scenes [12]. By introducing the adversarial adaptive clustering loss, cross-domain adaptative clustering (CDAC)

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networks are constructed for semi-supervised domain adaptation [13]. Unlike the two domains assumption of typical domain adaptation methods, the dynamic instance domain adaptation (DIDA) networks take each instance as a specific domain and combine it with cross-entropy loss to accomplish domain-adaptive recognition [14]. In addition, based on advanced deep learning frameworks, some deep networks are developed for cross-source object recognition. Using a transformer framework [15], the domain-specific representation networks are constructed using contrastive clustering loss for unsupervised domain adaptation [16]. With the aid of the strong representation power of diffusion model [15], the diffusion-based deep learning framework applies a denoising diffusion process to achieve effective object recognition results [17]. Combining with knowledge distillation technology, dual-head rectification domain adaptation network is developed using coarse-to-fine align strategy to learning domain-invariant features [18]. Although domain adaption deep learning methods have good cross-source recognition ability, the nature that they relay a large number of samples limits their scope of application. In contrast, domain adaption machine learning methods have a sound theoretical foundation, and their lightweight models give them an inherent performance advantage for small sample cases. Among machine learning methods, the most prominent representative is the support vector machine (SVM) [19], which has good recognition ability under small sample cases and thus has been integrated into remote sensing processing software (e.g., ENVI and EnMAP). To process cross-source remote sensing data, the conventional SVM was extended to domain adaptation versions, including A-SVM [20] and DA-SVM [21]. The former utilizes the decision function trained from samples of source 1 and the delta function trained from samples of source 2 to achieve object recognition, and the latter learns iteratively cross-source classification hyperplane parameters by continuously updating the weights of samples from different sources. To exploit the dependent relationship between data from different sources, a domain-dependent regularizer was absorbed into the SVM to achieve cross-source object recognition, making the nearby instances from different sources have similar projection values [22]. To improve classification power of linear projection, the domain transfer multiple kernel learning (DTMKL) method was developed to adjust the weights of different kernels so that the discrepancy between different sources can be reduced [23]. To excavate the latent domains from multisource data, the low-rank exemplar SVM was constructed to exploit the nuclear norm and adaptive weights of sources to obtain an effective classifier from the target domain [24]. Considering that data from different sources may present different dimensions, iterative reweight heterogeneous SVM was established, which first maps multisource samples to a feature space with the same dimension and then forms a classification hyperplane to classify cross-source samples [25]. Furthermore, assuming that the source and target domain samples share similar classification hyperplane parameters, the domain transfer SVM was constructed for multi-temporal remote sensing image classification [26]. To increase the recognition accuracy during the sequence sample acquisition, the incremental SVM was constructed by updating the solution of dual problem efficiently to improve the performance of the model [27].

The above domain adaptation SVM can be regarded as vector-based methods, whereas the remote sensing images can be naturally represented as tensor. When using vector-based methods to process remote sensing images, vectorization preprocessing has to be applied to make the data meet its input requirements, but this conversion easily leads to the loss of the intrinsic structural information of the remote sensing data and weakens the object recognition performance [18], [19]. Thus, the researchers developed the supervised tensor learning theory to extend the SVM to the tensor pattern to accept input represented as a tensor [30], e.g., C-support tensor machine (C-STM), ν − STM. In addition, by extending the rank-1 projection tensor used in C-STM to a higher rank, the higher-rank support tensor machine was constructed to achieve more effective object recognition [29]. To solve multiclass classification, the support multimodal tensor machine was generated to leverage multimodal product operations to process multiclass input [31]. To measure the importance of different samples, the weighted STM was built to weigh each sample and obtain better classification results [32]. Note that these variants are all designed for single-source data. When dealing with cross-source object recognition tasks, all of these methods need to be combined with feature transfer techniques, such as transfer component analysis (TCA) [33], active transfer learning (ATL) [34] method, and discriminative joint distribution adaptation (DJDA) [35], and probability-based graph embedding cross-domain and class discriminative adaptation (PGCD) [36], to transfer multisource data to the same feature space first. Then, the classifier can be trained to complete cross-source object recognitions in a step-by-step manner.

Although cross-source recognition can be performed indirectly by combining feature tensor transfer techniques with STM, a better way is to build a tensor-based domain adaptation classifier to separate cross-source tensor data directly. Considering the heterogeneity and dynamic nature of multisource satellite observation, there are several challenging problems in building a tensor-based domain adaptation classifier. First, the multisource remote sensing data have different resolutions and physical properties, which can be characterized as tensors with different dimensions and different distributions, denoted as heterogeneous tensors in this article. The classical STM can only handle the same-dimensional same-distributed homogeneous tensor but cannot adapt to the heterogeneous tensor remote sensing data with varied distribution and complex structure. Second, considering the cost of labeling the classes of training samples, some samples may lack the class labels, while existing STM can only learn hyperplane from labeled samples and cannot excavate the information from unlabeled samples. Third, to exploit the sequence samples obtained from multisource satellite observation, the domain adaptation STM is required to have the ability to update the model parameters online using the acquired sequence training samples to gradually improve the cross-source recognition performance. To conquer these challenges, the domain adaptation support tensor machine (DA-STM) method is proposed

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using a unified model to adapt effectively to the heterogeneous structure and diverse distribution of cross-source data to achieve cross-source heterogeneous data object recognition. To handle cross-source heterogeneous tensor data, the proposed DA-STM first constructs factor matrices with different modes to support the input of cross-source heterogeneous tensor data. To mine the shared information of cross-source data, the shared core tensor is constructed to learn coupledly the classification hyperplane of cross-source heterogeneous data in a unified feature space to separate cross-source objects adaptively. Moreover, by utilizing the adaptive sample labels, the hyperplane of DA-STM can be optimized by discriminant information from unlabeled samples. Furthermore, to utilize multisource sequence samples to improve the recognition performance continuously, the online version of DA-STM is further proposed to enable online updating of the models. To increase training efficiency to improve the practical application of the DA_STM, the DA-STM-oriented decomposition algorithm is presented to solve the dual problem of DA-STM efficiently. The main contributions are summarized as follows.

1) To recognize the objects described by cross-source remote sensing data with heterogeneous tensor structures and various data distributions, unlike the existing STM can only handle data with homogeneous structure and the same distribution, the DA-STM is proposed to learn different source-specific projection tensors constrained by the shared core tensor to deal with cross-source data, which breaks the basic assumption of the classical machine learning on the same structure and the same distribution of the input data, and ensures the effectiveness of cross-source remote sensing object recognition.
2) Compared with the standard STM, which can only be trained using offline samples, the online DA-STM is proposed to combine with the online learning mechanism to improve the cross-source recognition performance continuously using multiple source samples, making the model gradually adapt to the heterogeneous structure and complex distribution of cross-source data.
3) Considering that the DA-STM has complex dual problem compared with the STM, and the existing optimization solvers cannot solve this problem efficiently, the DA-STM-oriented decomposition algorithm is proposed to decompose the complex quadratic optimization into a series of simple analytic optimizations, thus accelerating the solving of parameters and ensuring the efficient implementation of DA-STM.

The remainder is organized as follows. Section II introduces the preliminaries, including the basic tensor algebra and STM method. In Section III, the proposed DA-STM and the corresponding solving method are given. In Section IV, the proposed online version of DA-STM is proposed, and the relationship between existing methods and DA-STM is analyzed. In Section V, the experiments are conducted on multiresolution multi-angle remote sensing dataset and multi-angle SAR dataset to evaluate the performance of the proposed DA-STM compared with typical domain adaptation methods. Our conclusion is given in Section VI.

II. Guidelines for Manuscript Preparation

Before presenting the DA-STM, the notations, basic tensor operators, and standard STM are introduced in succession, facilitating the understanding of the proposed DA-STM.

A. Notations and Basic Tensor Operators

Following the convention used in [37], we use scalar, vector, matrix, and tensor to denote an italic lowercase letter, e.g., x and y, lowercase boldface letter, e.g., $\boldsymbol{x}$ and $\boldsymbol{y}$, uppercase boldface letter, e.g., $\boldsymbol{M}$, and calligraphic letter, e.g., $\mathcal{X}$, respectively. In addition, for convenience of indexing entries of tensor, the MATLAB notation $\mathcal{X}(i, j, k)$ is adopted to denote the value of tensor in entry (i, j, k). Then, the tensor operation used throughout this article is given as follows.

Definition 1 (Tucker Decomposition of a Tensor): Given a tensor $\mathcal{X} \in \mathbb{R}^{I_1,\ldots,I_N}$, the Tucker decomposition [39] of a tensor defines an approximation representation of $\mathcal{X}$, i.e., $\mathcal{X} \approx \mathcal{G} \times_1 \boldsymbol{U}_1 \times_2 \boldsymbol{U}_2 \cdots \times_N \boldsymbol{U}_N$, where $\mathcal{G} \in \mathbb{R}^{i_1,\ldots,i_N}$ and $\boldsymbol{U}_1 \in \mathbb{R}^{I_1,i_1}$ denote core tensor and factor matrix, respectively.

Definition 2 (Frobenius Norm of a Tensor): Given a tensor $\mathcal{A} \in \mathbb{R}^{I_1,\ldots,I_M}$, the Frobenius norm is defined as

$$\|\mathcal{A}\|_F = \sqrt{\sum_{i_1,\ldots,i_M} |\mathcal{A}(i_1, \ldots, i_M)|^2}. \tag{1}$$

Definition 3 (Tensor-Matrix Product): The tensor-matrix product between tensor $\mathcal{A} \in \mathbb{R}^{I_1,\ldots,I_M}$ and matrix $\boldsymbol{B} \in \mathbb{R}^{I'_k,I_k}$ along with mode-k is denoted as $\mathcal{C} = \mathcal{A} \times_k \boldsymbol{B}$, where $\mathcal{C} \in \mathbb{R}^{I_1,\ldots,I'_k,\ldots,I_M}$ denotes the resulting tensor, and the entry $(I_1, \ldots, I_{k-1}, i'_k, I_{k+1}, \ldots, I_M)$ of $\mathcal{C}$ can be obtained as

$$\sum_{i_k=1}^{I_k} \mathcal{A}(I_1, \ldots, I_{k-1}, i_k, I_{k+1}, \ldots, I_M) \times \boldsymbol{B}\left(i'_k, i_k\right). \tag{2}$$

Definition 4 (Tensor Unfolding): The matrixization unfolding of a tensor $\mathcal{A} \in \mathbb{R}^{I_1,\ldots,I_M}$ along with mode-k is a matrix denoted by $\mathrm{Mat}_{(k)}(\mathcal{A}) \in \mathbb{R}^{I_k, \prod_{i\neq k} I_i}$, where the entry $(i_k, \sum_j i_j \times \prod_{i\neq k}^{j-1} I_i)$ of $\mathrm{Mat}_{(k)}(\mathcal{A})$ is the value of $\mathcal{A}(i_1, \ldots, i_M)$. In addition, the vectorization of $\mathcal{A} \in \mathbb{R}^{I_1,\ldots,I_M}$ is a vector denoted by $\mathrm{vec}(\mathcal{A}) \in \mathbb{R}^{\prod_{l=1}^{M} I_l}$, where the entry $(\sum_j i_j \times \prod_{i=1}^{j-1} I_i)$ of $\mathrm{vec}(\mathcal{A})$ is the value of $\mathcal{A}(i_1, \ldots, i_M)$.

Definition 5 (Inner Product of Tensors): Given two tensors $\mathcal{A} \in \mathbb{R}^{I_1,\ldots,I_M}$ and $\mathcal{B} \in \mathbb{R}^{I_1,\ldots,I_M}$ of the same size, the inner product operator contract these two tensors to a scalar calculated by $\langle \mathcal{A}, \mathcal{B} \rangle = \sum_{i_1=1,\ldots,i_M=1}^{i_1=I_1,\ldots,i_M=I_M} \mathcal{A}(i_1, \ldots, i_M) \times \mathcal{B}(i_1, \ldots, i_M)$.

Definition 6 (Outer Product): Given a group of vectors $\{\boldsymbol{w}_l \in \mathbb{R}^{I_l}\}_{l=1}^{M}$, the outer product (i.e., tensor product) of these vectors $\boldsymbol{w}_1 \circ \cdots \circ \boldsymbol{w}_M$ is a M-order tensor $\mathcal{W}$ with dimensions $I_1, \ldots, I_M$, and the entry of tensor is calculated by $\mathcal{W}(i_1, \ldots, i_M) = \boldsymbol{w}_1(i_1) \times \cdots \times \boldsymbol{w}_M(i_M)$.

B. Support Tensor Machine

When processing remote sensing data represented as M-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times,\ldots,\times I_M}$, the vectorization preprocessing has to be employed by SVM to convert the input to a vector $\boldsymbol{x} \in \mathbb{R}^{\prod_j I_j}$. This conversion inevitably causes the loss of tensor structure information and a decrease in recognition results. To maintain the tensor structure of input data, the STM

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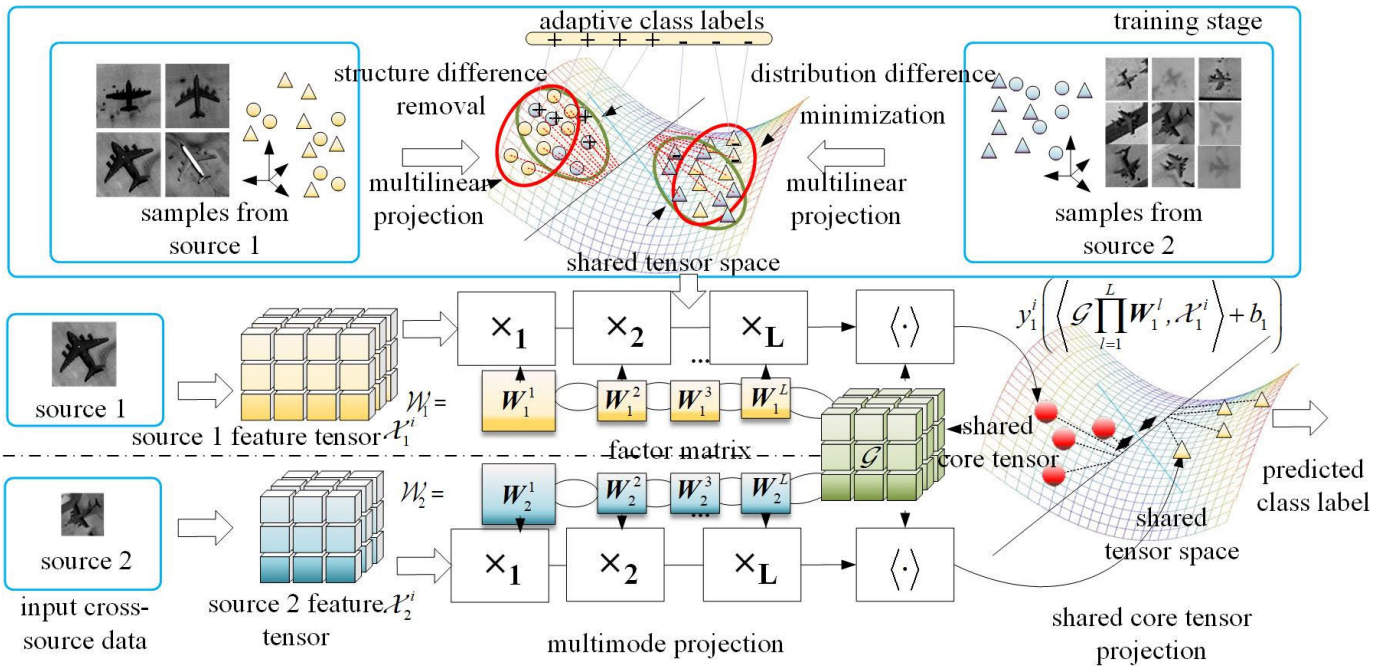


Fig. 1. Illustration of the cross-source object recognition procedure using the proposed DA-STM.

is extended from SVM to separate directly input in the tensor space, as

$$y(\mathcal{X}) = \text{sign}\left(\mathcal{X}\prod_{j=1}^{M}\times_j \boldsymbol{w}_j + b\right) \tag{3}$$

where $\boldsymbol{w}_j$ and b denote the projection vector of jth mode and the bias, respectively. To learn optimal $\boldsymbol{w}_j$, given training set $\{\mathcal{X}_i, y_i\}|_{i=1}^{N}$, where $\mathcal{X}_i \in \mathbb{R}^{I_1\times\cdots\times I_M}$ and $y_i \in \{1, -1\}$ denote the feature tensor of ith sample and the class label, respectively, the parameter model of STM can be obtained by solving the following optimization:

$$\begin{aligned}&\min_{\boldsymbol{w}_i|_{i=1}^{M},b}\ \frac{1}{2}\|\boldsymbol{w}_1\circ\cdots\circ\boldsymbol{w}_M\|_F^2 + C\sum_{i=1}^{N}\xi_i\\ &\text{s.t. } y_i\left(\mathcal{X}_i\prod_{j=1}^{M}\times_j\boldsymbol{w}_j + b\right)\geq 1-\xi_i\\ &\xi_i\geq 0,\quad 1\leq i\leq N\end{aligned} \tag{4}$$

where ξ_i and C denote the slacking factor and regularization factor, respectively. The classification hyperplane is characterized by $\boldsymbol{w}_1\circ\cdots\circ\boldsymbol{w}_M$. After using the alternative optimization strategy, the optimal $\boldsymbol{w}_i|_{i=1}^{M}$ and b can be obtained for object recognition.

III. Domain Adaptation Support Tensor Machine

Note that the classical STM can only handle single-source homogeneous tensor data and fail to deal with cross-source heterogeneous tensor data. To efficiently recognize the objects described by cross-source remote sensing data, the DA-STM method is proposed, and its overall procedure is illustrated in Fig. 1. For the training stage, the cross-source slices are used to learn multiple heterogeneous projection tensors constrained by the shared core tensor to reduce the structure discrepancy and distribution difference across sources. For the test stage, the slice from any source is represented as a feature tensor, and the trained cross-source projection tensor of DA-STM is then applied to separate feature tensor in tensor space.

For ease of understanding and without loss of generality, consider the case of two sources, and source 2 contains some unlabeled samples. Given samples from source 1 $\{\mathcal{X}_1^n, y_1^n\}_{n=1}^{N_1}$, labeled samples from source 2 $\{\mathcal{X}_2^n, y_2^n\}_{n=1}^{N_2}$, and unlabeled samples from source 2 $\{\mathcal{X}_2^n, y_2^n\}_{n=N_2+1}^{N_2+N_{2_ul}}$, where $\mathcal{X}_1^n \in \mathbb{R}^{I_1,\ldots,I_L}$, $y_1^n \in \{-1, 1\}$, $\mathcal{X}_2^n \in \mathbb{R}^{I_1',\ldots,I_L'}$, $y_2^n \in \{-1, 1\}$ denote the nth sample from source 1, the nth sample label of source 1, nth sample from source 2, the nth sample label of source 2, respectively. Since source 1 and 2 data are observed from different sensors, they may present different spatial or spectral resolutions, and they thus can be represented as tensors with different dimensions (i.e., heterogeneous tensors). Thus, we allow $I_l \neq I_l'$, $1 \leq l \leq L$. Additionally, data from different sources may present diverse distribution, i.e., $\mathcal{D}(\mathcal{X}_1^n) \neq \mathcal{D}(\mathcal{X}_2^n)$. In this way, the purpose of DA-STM is to construct cross-source shared classification hyperplanes for object classification of arbitrary samples from source 1 or source 2.

The classical STM performs recognition to express the input as a single tensor with a fixed size. For cross-source data, it is necessary to build multiple projection tensors with different dimensions consistent with the input to support cross-source heterogeneous tensors, i.e., construct $\mathcal{W}_1 \in \mathbb{R}^{I_1,\ldots,I_L}$ and $\mathcal{W}_2 \in \mathbb{R}^{I_1',\ldots,I_L'}$. Furthermore, compared with the standard STM that limits the projection tensor as a rank-1 tensor $\mathcal{W} = \boldsymbol{w}_1\circ\boldsymbol{w}_2\circ\cdots\circ\boldsymbol{w}_L$, according to Tucker decomposition, the heterogeneous projection tensors in DA-STM are defined as high-rank tensors, i.e., $\mathcal{W}_1 = \mathcal{G}_1\times_1\boldsymbol{W}_1^1\times_2\cdots\times_L\boldsymbol{W}_1^L$ and $\mathcal{W}_2 = \mathcal{G}_2\times_1\boldsymbol{W}_2^1\times_2\cdots\times_L\boldsymbol{W}_2^L$, where $\mathcal{G}_1$ and $\mathcal{G}_2$ denote core tensors, and $\boldsymbol{W}_1^1$ and $\boldsymbol{W}_2^1$ denote the factor matrices. Compared with existing STM trained independently for different sources,

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the key for cross-source object recognition is to explore the shared information of multisource data to enhance object recognition performance. To effectively capture the shared information of multisource data, the shared core tensor is established to learn the shared classification hyperplane to cope with the cross-source heterogeneous data. Specifically, let $\mathcal{G}_1 = \mathcal{G}_2 = \mathcal{G}$, where $\mathcal{G} \in \mathbb{R}^{i_1, i_2, \ldots, i_L}$ denotes the shared core tensor used for both two sources. Then terms $i_1, \ldots, i_L$ indicate the dimensions of the shared core tensor.

For the multisource samples $\mathcal{X}_1^n$ and $\mathcal{X}_2^n$, the inner product between $\mathcal{W}_1$ ($\mathcal{W}_2$) and $\mathcal{X}_1^n$ ($\mathcal{X}_2^n$) can be expressed as $\langle \mathcal{W}_1, \mathcal{X}_1^n \rangle$ and $\langle \mathcal{W}_2, \mathcal{X}_2^n \rangle$. Then, combining with the high-rank tensor expression of Tucker decomposition, we have $\langle \mathcal{G} \times_1 \boldsymbol{W}_1^1 \times_2 \cdots \times_L \boldsymbol{W}_1^L, \mathcal{X}_1^n \rangle$ and $\langle \mathcal{G} \times_1 \boldsymbol{W}_2^1 \times_2 \cdots \times_L \boldsymbol{W}_2^L, \mathcal{X}_2^n \rangle$. It can be simplified as:

$$\left\langle \mathcal{G} \times_1 \boldsymbol{W}_1^1 \times_2 \cdots \times_L \boldsymbol{W}_1^L, \mathcal{X}_1^n \right\rangle = \left\langle \mathcal{G}, \mathcal{X}_1^n \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^{lT} \right\rangle \quad (5)$$

$$\left\langle \mathcal{G} \times_1 \boldsymbol{W}_2^1 \times_2 \cdots \times_L \boldsymbol{W}_2^L, \mathcal{X}_2^n \right\rangle = \left\langle \mathcal{G}, \mathcal{X}_2^n \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^{lT} \right\rangle. \quad (6)$$

The above expression can be understood from another perspective. The original features $\mathcal{X}_1^n$ and $\mathcal{X}_2^n$ are mapped into an intermediate sub-space using factor matrices to eliminate the structure difference between different sources. Then, apply the shared core tensor $\mathcal{G}$ as a projection tensor to separate the cross-source data into class space.

Additionally, considering that the training set contains unlabeled samples, it is necessary for the DA-STM to autonomously mine the potential category labels of unlabeled samples. To obtain this ability, the adaptive class labels are constructed as the variables to be optimized to learn the label information of the unlabeled samples. Integrating the factor matrices, shared core tensor, and adaptive class labels to standard STM, we get the optimization problem of DA-STM:

$$\begin{aligned}
\min_{\mathcal{G}, \boldsymbol{W}_1^l, \boldsymbol{W}_2^l, b_1, b_2, y_2^i|_{i=N_2+1}^{N_2+N_{2_ul}}} & \frac{1}{2}\left(\left\| \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l \right\|_F^2 + \left\| \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l \right\|_F^2 \right) \\
& + C_1 \sum_{i=1}^{N_1} \xi_1^i + C_2 \sum_{i=1}^{N_2} \xi_2^i + C_3 \sum_{i=N_2+1}^{N_2+N_{2_ul}} \xi_2^i \\
\text{s.t. } & y_1^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l, \mathcal{X}_1^i \right\rangle + b_1 \right) \geq 1 - \xi_1^i \\
& \xi_1^i \geq 0, \quad 1 \leq i \leq N_1 \\
& y_2^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i \right\rangle + b_2 \right) \geq 1 - \xi_2^i \\
& \xi_2^i \geq 0, \quad 1 \leq i \leq N_2 + N_{2_ul} \\
& y_2^i \in \{-1, 1\}, \quad \sum_i y_{2_ul}^i = N' \\
& N_2 + 1 \leq i \leq N_2 + N_{2_ul}
\end{aligned} \quad (7)$$

where ξ_1^i, ξ_2^i, C_1, C_2, C_3, and $y_2^i|_{i=N_2+1}^{N_2+N_{2_ul}}$ denote the slacking factors of source 1, slacking factors of source 2, the regularization factor of labeled samples from source 1, the regularization factor of labeled samples from source 2, the regularization factor of unlabeled samples from source 2, and the adaptive class labels, respectively. The N' is a pre-defined const used to indicate the distribution of unlabeled samples class, which is beneficial for adaptive class labels to learn more accurate labels during the optimization of model.

A. Solving the Optimization Problem of DA-STM

It is mentioned that the optimization problem of DA-STM is non-convex, and thus the alternative optimization scheme is taken into consideration to solve this problem, i.e., optimize the factor matrices, shared core tensor, and adaptive sample labels in turn until the optimization problem converges.

1) Update the $\boldsymbol{W}_1^l$: When optimizing the variable $\boldsymbol{W}_1^l$, the other factor matrices and shared core tensor are fixed. Then, the current sub-optimization is derived as

$$\begin{aligned}
\min_{\boldsymbol{W}_1^l, b_1} & \frac{1}{2} \left\| \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l \right\|_F^2 + C_1 \sum_{i=1}^{N_1} \xi_1^i \\
\text{s.t. } & y_1^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l, \mathcal{X}_1^i \right\rangle + b_1 \right) \geq 1 - \xi_1^i \\
& \xi_1^i \geq 0, \quad 1 \leq i \leq N_1.
\end{aligned} \quad (8)$$

For ease of solving, let $\widetilde{\boldsymbol{W}}_1^l = \boldsymbol{W}_1^l \boldsymbol{K}^{(1/2)}$, where $\boldsymbol{K}$ is defined by:

$$\boldsymbol{K} = \mathrm{Mat}_{(l)} \left(\mathcal{G} \prod_{j \neq l} \times_j \boldsymbol{W}_1^j \right) \mathrm{Mat}_{(l)}^T \left(\mathcal{G} \prod_{j \neq l} \times_j \boldsymbol{W}_1^j \right). \quad (9)$$

Then, let $\widetilde{\boldsymbol{X}_i} = \boldsymbol{K}^{-(1/2)} \times \mathrm{Mat}_{(l)}(\mathcal{G} \prod_{j \neq l} \times_j \boldsymbol{W}_1^j) \mathrm{Mat}_{(l)}^T(\mathcal{X}_1^i)$. Note that the $\boldsymbol{K}$ may be singular, and the $\boldsymbol{K}^{-(1/2)}$ can be calculated by $(\boldsymbol{K} + \varepsilon \boldsymbol{I})^{-(1/2)}$, where ε denotes a small positive number. The corresponding intermediator variable can be permuted as a vector, i.e., $\widetilde{\boldsymbol{w}}_1^l = \mathrm{vec}(\widetilde{\boldsymbol{W}}_1^l)$, $\widetilde{\boldsymbol{x}_i} = \mathrm{vec}(\widetilde{\boldsymbol{X}_i})$. Substitute these variables to (8), we have

$$\begin{aligned}
\min_{\widetilde{\boldsymbol{w}}_1^l, b_1} & \frac{1}{2} \widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{w}}_1^l + C_1 \sum_{i=1}^{N_1} \xi_1^i \\
\text{s.t. } & y_1^i \left(\widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{x}}_i + b_1 \right) \geq 1 - \xi_1^i \\
& \xi_1^i \geq 0, \quad 1 \leq i \leq N_1.
\end{aligned} \quad (10)$$

It is noted that the representation of the optimization problem is greatly simplified by variable substitution. To eliminate the inequation constraints, the Lagrangian function is drawn as:

$$\begin{aligned}
\mathcal{L}\left(\widetilde{\boldsymbol{w}}_1^l, b_1, \xi_1^i, \alpha_i, \lambda_i \right) = & \frac{1}{2} \widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{w}}_1^l + C_1 \sum_{i=1}^{N_1} \xi_1^i \\
& - \sum_{i=1}^{N_1} \alpha_i \left(y_1^i \left(\widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{x}}_i + b_1 \right) - 1 + \xi_1^i \right) - \sum_{i=1}^{N_1} \lambda_i \xi_1^i
\end{aligned} \quad (11)$$

where α_i, λ_i denote the Lagrangian multiplier. Let the partial derivative of (11) with respect to $\boldsymbol{W}_1^l$, b, ξ_1^i be zeros. We have:

$$\frac{\partial \mathcal{L}}{\partial \widetilde{\boldsymbol{w}}_1^l} = 0 \Rightarrow \widetilde{\boldsymbol{w}}_1^l = \sum_{i=1}^{N_1} \alpha_i y_1^i \widetilde{\boldsymbol{x}}_i$$

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$$\frac{\partial \mathcal{L}}{\partial b_1} = 0 \Rightarrow \sum_{i=1}^{N_1} y_1^i \alpha_i = 0$$
$$\frac{\partial \mathcal{L}}{\partial \xi_1^i} = 0 \Rightarrow \alpha_i + \lambda_i = C_1. \tag{12}$$

In addition, substitute the above equations to (8). The obtained Lagrangian dual problem is shown below:

$$\begin{aligned} &\min_{\boldsymbol{\alpha}} \boldsymbol{\alpha}^T \boldsymbol{K} \boldsymbol{\alpha} - \boldsymbol{e}^T \boldsymbol{\alpha} \\ &\text{s.t.} \sum_{i=1}^{N_1} y_1^i \alpha_i = 0 \\ &\quad 0 \le \alpha_i \le C_1, \quad 1 \le i \le N_1 \end{aligned} \tag{13}$$

where $\boldsymbol{K}(i,k) = (1/2) y_1^i y_1^k \widetilde{\boldsymbol{x}}_i^T \widetilde{\boldsymbol{x}}_k$. The above optimization is quadratic programming and can be solved via the classical interior-point algorithm. The more efficient solving method can be found in Section III-C. For the optimized α_i, (12) can be utilized to obtain the optimized $\widetilde{\boldsymbol{w}}_1^l$. Then, apply $\boldsymbol{W}_1^l = \widetilde{\boldsymbol{W}}_1^l \boldsymbol{K}^{-(1/2)}$ to obtain the optimized factor matrix.

2) Update the $\boldsymbol{W}_2^l$: In sequence, the $\boldsymbol{W}_2^l$ will be updated. This process is similar to the updating of $\boldsymbol{W}_1^l$, and the details can be found in the Appendix.

3) Update the Shared Core Tensor: When optimizing the shared core tensor, the remained variables are fixed. The current sub-optimization is:

$$\begin{aligned} \min_{\mathcal{G}, b_1, b_2} \ & \frac{1}{2} \left(\left\| \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l \right\|_F^2 + \left\| \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l \right\|_F^2 \right) \\ & + C_1 \sum_{i=1}^{N_1} \xi_1^i + C_2 \sum_{i=1}^{N_2} \xi_2^i + C_3 \sum_{i=N_2+1}^{N_2+N_{2_ul}} \xi_2^i \\ \text{s.t.} \ & y_1^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l, \mathcal{X}_1^i \right\rangle + b_1 \right) \ge 1 - \xi_1^i \\ & \xi_1^i \ge 0, \quad 1 \le i \le N_1 \\ & y_2^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i \right\rangle + b_2 \right) \ge 1 - \xi_2^i \\ & \xi_2^i \ge 0, \quad 1 \le i \le N_2 + N_{2_ul}. \end{aligned} \tag{14}$$

For ease of solving, use the variables substitution:

$$\begin{aligned} &\boldsymbol{K} = \sum_i \left(\boldsymbol{W}_i^1 \otimes \cdots \otimes \boldsymbol{W}_i^L \right) \left(\boldsymbol{W}_i^1 \otimes \cdots \otimes \boldsymbol{W}_i^L \right)^T \\ &\widetilde{\boldsymbol{g}} = \mathrm{vec}(\mathcal{G}) \boldsymbol{K}^{\frac{1}{2}}, \quad \widetilde{\boldsymbol{x}_1^i} = \boldsymbol{K}^{-\frac{1}{2}} \mathrm{vec} \left(\mathcal{X}_1^i \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l \right) \\ &\widetilde{\boldsymbol{x}_2^i} = \boldsymbol{K}^{-\frac{1}{2}} \mathrm{vec} \left(\mathcal{X}_2^i \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l \right). \end{aligned} \tag{15}$$

Bringing the intermediate variables into the sub-optimization, we have:

$$\begin{aligned} \min_{\widetilde{\boldsymbol{g}}, b_1, b_2} \ & \frac{1}{2} \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{g}} + C_1 \sum_{i=1}^{N_1} \xi_1^i + C_2 \sum_{i=1}^{N_2} \xi_2^i + C_3 \sum_{i=N_2+1}^{N_2+N_{2_ul}} \xi_2^i \\ \text{s.t.} \ & y_1^i \left(\widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{x}}_1^i + b_1 \right) \ge 1 - \xi_1^i, \quad \xi_1^i \ge 0, \ 1 \le i \le N_1 \\ & y_2^i \left(\widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{x}}_2^i + b_2 \right) \ge 1 - \xi_2^i \\ & \xi_2^i \ge 0, \quad 1 \le i \le N_2 + N_{2_ul}. \end{aligned} \tag{16}$$

Then, draw the Lagrangian function as

$$\begin{aligned} &\mathcal{L}\left(\widetilde{\boldsymbol{g}}, b_1, b_2, \xi_1^i, \xi_2^i, \alpha_i^1, \alpha_i^2, \lambda_i^1, \lambda_i^2\right) \\ &= \frac{1}{2} \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{g}} + C_1 \sum_{i=1}^{N_1} \xi_1^i + C_2 \sum_{i=1}^{N_2} \xi_2^i + C_3 \sum_{i=N_2+1}^{N_2+N_{2_ul}} \xi_2^i \\ &\quad - \sum_{i=1}^{N_1} \alpha_i^1 \left(y_1^i \left(\widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{x}}_1^i + b_1 \right) - 1 + \xi_1^i \right) \\ &\quad - \sum_{i=1}^{N_2+N_{2_ul}} \alpha_i^2 \left(y_1^i \left(\widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{x}}_2^i + b_2 \right) - 1 + \xi_2^i \right) \\ &\quad - \sum_{i=1}^{N_1} \lambda_i^1 \xi_1^i - \sum_{i=1}^{N_2+N_{2_ul}} \lambda_i^2 \xi_2^i \end{aligned} \tag{17}$$

where $\alpha_i^1, \alpha_i^2, \lambda_i^1, \lambda_i^2$ denote the dual variables. Let the partial derivatives of the above function respect to $\mathcal{G}, b_1, b_2, \xi_1^i, \xi_2^i$ be zeros. Then

$$\begin{aligned} &\frac{\partial \mathcal{L}}{\partial \widetilde{\boldsymbol{g}}} = 0 \Rightarrow \\ &\quad \widetilde{\boldsymbol{g}} = \sum_{i=1}^{N_1} \alpha_i^1 y_1^i \widetilde{\boldsymbol{x}}_1^i + \sum_{i=1}^{N_2+N_{2_ul}} \alpha_i^2 y_2^i \widetilde{\boldsymbol{x}}_2^i \\ &\frac{\partial \mathcal{L}}{\partial b_1} = 0 \Rightarrow \sum_{i=1}^{N_1} y_1^i \alpha_i^1 = 0 \\ &\frac{\partial \mathcal{L}}{\partial b_2} = 0 \Rightarrow \sum_{i=1}^{N_2+N_{2_ul}} y_2^i \alpha_i^2 = 0 \\ &\frac{\partial \mathcal{L}}{\partial \xi_1^i} = 0 \Rightarrow \alpha_i^1 + \lambda_i^1 = C_1 \\ &\frac{\partial \mathcal{L}}{\partial \xi_2^i} = 0 \Rightarrow \alpha_i^2 + \lambda_i^2 = C_2, \quad 1 \le i \le N_2 \\ &\alpha_{N_2+i}^2 + \lambda_{N_2+i}^2 = C_3, \quad 1 \le i \le N_{2_ul}. \end{aligned} \tag{18}$$

Substituting above into (17) yields the dual problem:

$$\begin{aligned} &\min_{\boldsymbol{\alpha}} \boldsymbol{\alpha}^T \boldsymbol{K} \boldsymbol{\alpha} - \boldsymbol{e}^T \boldsymbol{\alpha} \\ &\text{s.t.} \sum_{i=1}^{N_1} y_1^i \alpha_i^1 = 0, \quad 1 \le i \le N_1 \\ &\quad \sum_{i=1}^{N_1} y_2^i \alpha_i^2 = 0, \quad 1 \le i \le N_2 \\ &\quad 0 \le \alpha_i^1 \le C_1, \quad 1 \le i \le N_1 \\ &\quad 0 \le \alpha_i^2 \le C_2, \quad 1 \le i \le N_2 \\ &\quad 0 \le \alpha_{i+N_2}^2 \le C_3, \quad 1 \le i \le N_{2_ul} \end{aligned} \tag{19}$$

where $\boldsymbol{\alpha} = [\alpha_1^1, \ldots, \alpha_{N_1}^1, \alpha_1^2, \ldots, \alpha_{N_2+N_{2_ul}}^2]$. The $\boldsymbol{e}$ is the summing vector with all elements of 1. The entry of $\boldsymbol{K}$ is equal to:

$$\begin{aligned} &\boldsymbol{K}((g-1) \times N_1 + i, (h-1) \times N_1 + k) \\ &\qquad = \frac{1}{2} \left\langle \mathrm{vec}\left(y_g^i \widetilde{\boldsymbol{x}}_g^i \right), \mathrm{vec}\left(y_h^k \widetilde{\boldsymbol{x}}_h^k \right) \right\rangle. \end{aligned} \tag{20}$$

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Apply the quadratic programming solver or the decomposition algorithm presented in Section III-B, the optimal $\boldsymbol{\alpha}$ can be obtained. Then, use $\widetilde{\boldsymbol{g}} = \sum_{i=1}^{N_1} \alpha_i^1 y_1^i \widetilde{\boldsymbol{x}}_1^i + \sum_{i=1}^{N_2+N_{2_ul}} \alpha_i^2 y_2^i \widetilde{\boldsymbol{x}}_2^i$ to update $\widetilde{\boldsymbol{g}}$, and utilize $\text{vec}(\mathcal{G}) = \widetilde{\boldsymbol{g}} \boldsymbol{K}^{-(1/2)}$ to calculate the updated shared core tensor $\mathcal{G}$.

4) Update the Adaptive Class Labels: When optimizing the adaptive class labels $y_i^2, N_2 + 1 \le i \le N_2 + N_{2_ul}$, the corresponding sub-optimization is shown below:

$$\begin{aligned} &\min_{y_2^i} \max \left(0, 1 - y_2^i \left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i \right\rangle + b_2\right)\right) \\ &\text{s.t. } y_2^i \in \{-1, 1\}, N_2 + 1 \le i \le N_2 + N_{2_ul} \\ &\quad \#\left\{y_2^i | y_2^i = 1, N_2 + 1 \le i \le N_2 + N_{2_ul}\right\} = N_{2_ul}^+. \end{aligned} \tag{21}$$

The operator $\#\{\cdot\}$ denotes the number of elements in the set. The $N_{2_ul}^+ = ((N_{2_ul} + N')/2)$ indicates the number of positive samples in the unlabeled sample set. The above optimization belongs to combinatorial optimization. To facilitate solving, the above equation is transformed via convex relaxation strategy into the following linear programming problem:

$$\begin{aligned} &\min_{\boldsymbol{Y}} \text{tr}\left(\boldsymbol{V}^T \boldsymbol{Y}\right) \\ &\text{s.t. } \sum_{j=1}^{2} \boldsymbol{Y}(i, j) = 1, \sum_{i=1}^{N_{2_ul}} \boldsymbol{Y}(i, 1) = N_{2_ul}^+ \\ &\quad \boldsymbol{Y}(i, j) \ge 0 \end{aligned} \tag{22}$$

where $\boldsymbol{V}(i, 1) = \max(0, 1 - (\langle \mathcal{G} \prod_{l=1}^{L} \mathcal{X}_l \boldsymbol{W}_2^l, \mathcal{X}_2^i \rangle + b_2))$ and $\boldsymbol{V}(i, 2) = \max(0, 1 + (\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i \rangle + b_2))$. The variables $\boldsymbol{Y} \in \mathbb{R}^{N_{2_ul}, 2}$ are the matrix form of adaptive class labels, where ith row of $\boldsymbol{Y}$ indicates the label of ith unlabeled sample. Note that the suboptimization after convex relaxation can be considered a linear transportation problem, which can be solved using the transportation simplex method [38]. When obtaining $\boldsymbol{Y}$, the largest column in ith row is selected to determine the category labels, i.e., the optimized adaptive labeling variables are calculated according to the following equation:

$$y_2^{N_2+i} = \begin{cases} 1, & \text{if } \arg\max_j \boldsymbol{V}(i, j) = 1 \\ -1, & \text{otherwise.} \end{cases} \tag{23}$$

Alternately update the factor matrix, core tensor, and adaptive class labels until the terminal conditions are met:

$$\left\|\mathcal{W}_1 - \mathcal{W}_1'\right\|_F^2 + \left\|\mathcal{W}_2 - \mathcal{W}_2'\right\|_F^2 < \delta || k > k_{\text{th}} \tag{24}$$

where $\mathcal{W}_1'$, $\mathcal{W}_2'$, δ, k, and k_{th} denote the heterogeneous projection tensors of source 1 updated in the last round, the heterogeneous projection tensors of source 2 updated in the last round, the pre-defined tolerance, the number of iterations, and the threshold of iteration number, respectively. When the alternative optimization method converges, combining with Karush–Kuhn–Tucker (K.K.T.) condition, the bias b_1 and b_2 can be calculated by

$$\begin{aligned} b_1 = -0.5 &\times \left(\frac{1}{\#\left\{i | y_1^i = 1, 0 < \alpha_1^i < C_1\right\}}\right. \\ &\times \sum_{\left\{i | y_1^i = 1, 0 < \alpha_1^i < C_1\right\}} \left(\mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l, \mathcal{X}_1^i\right) \\ &+ \frac{1}{\#\left\{i | y_1^i = -1, 0 < \alpha_1^i < C_1\right\}} \\ &\left.\times \sum_{\left\{i | y_1^i = -1, 0 < \alpha_1^i < C_1\right\}} \left(\mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l, \mathcal{X}_1^i\right)\right) \\ b_2 = -0.5 &\times \left(\frac{1}{\#\left\{i | y_2^i = 1, 0 < \alpha_2^i < C_2\right\}}\right. \\ &\times \sum_{\left\{i | y_2^i = 1, 0 < \alpha_2^i < C_2\right\}} \left(\mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i\right) \\ &+ \frac{1}{\#\left\{i | y_2^i = -1, 0 < \alpha_2^i < C_2\right\}} \\ &\left.\times \sum_{\left\{i | y_2^i = -1, 0 < \alpha_1^i < C_2\right\}} \left(\mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i\right)\right). \end{aligned} \tag{25}$$

When the training is finished, the DA-STM can be applied to recognize any sample from source 1 or source 2 using the following decision function:

$$\text{sgn}\left(\left\langle \mathcal{G} \prod_{l=1}^{L} \times_l \boldsymbol{W}_m^l, \mathcal{X}_m \right\rangle + b_m\right) \tag{26}$$

where $m \in \{1, 2\}$ denotes the index number of source for the sample to be recognized. The $\text{sgn}(\cdot)$ represents the sign function.

B. Accelerating the Training of DA-STM

The alternative optimization scheme of DA-STM shows that the dual problems must be solved alternatively to obtain the optimal model parameter. Applying the classical quadratic programming method to solve the dual problem will increase the time consumption due to their complex numerical iteration. To accelerate the training of DA-STM, one idea worth considering is to build efficient algorithms for solving the dual problem. Similar to sequential minimal optimization (SMO) used for SVM solving [39], the DA-STM-oriented decomposition algorithm is proposed to convert the complex numerical optimization of dual problem to a series of analytic optimization to accelerate the training of DA-STM.

For DA-STM, it is needed to solve three types of dual problems for updating $\boldsymbol{W}_1^l$, $\boldsymbol{W}_2^l$, and $\mathcal{G}$, respectively. Since the dual problems for updating $\boldsymbol{W}_1^l$ and $\boldsymbol{W}_2^l$ can be regarded as a particular case of the dual problem for updating $\mathcal{G}$, we take the dual problem with optimized variables $\mathcal{G}$ as an example to propose a DA-STM oriented decomposition algorithm, and the other two types of dual problems can be solved through the similar manners.

To solve the dual problem for updating $\mathcal{G}$ [see Eq. (19)], the $\boldsymbol{\alpha}$ is initialized to zero vector as an initial feasible solution. Unlike the interior-point method that optimizes all the variables simultaneously, the proposed decomposition algorithm

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divides the variables into a working set and a non-working set. In each round, only update the variables in the working set by fixing the variables in the non-working set. When only considering the working set variables, Eq. (19) is changed to the following form:

$$\begin{aligned}
&\min_{\boldsymbol{\alpha}_B} \boldsymbol{f}(\boldsymbol{\alpha}_B) = \boldsymbol{\alpha}_B^T K_{BB}\boldsymbol{\alpha}_B + 2\times\boldsymbol{\alpha}_B^T K_{BN}\boldsymbol{\alpha}_N - e^T\boldsymbol{\alpha}_B \\
&\text{s.t. } y_B^{1T}\boldsymbol{\alpha}_B^1 = -y_N^{1T}\boldsymbol{\alpha}_B^1 \\
&\quad y_B^{2T}\boldsymbol{\alpha}_B^2 = -y_N^{2T}\boldsymbol{\alpha}_B^2 \\
&\quad 0 \le \alpha_i^1 \le C\left(\alpha_i^1\right), \quad 1 \le i \le N_1 \\
&\quad 0 \le \alpha_i^2 \le C\left(\alpha_i^2\right), \quad 1 \le i \le N_2 + N_{2_ul}
\end{aligned} \tag{27}$$

where $\boldsymbol{\alpha}_B = [\boldsymbol{\alpha}_B^1, \boldsymbol{\alpha}_B^2]$ denotes the variables in working set. The $\boldsymbol{\alpha}_B^1$ with corresponding labels $\boldsymbol{y}_B^1$ is selected from $\boldsymbol{\alpha}^1$, and $\boldsymbol{\alpha}_B^2$ with corresponding labels $\boldsymbol{y}_B^2$ is selected from $\boldsymbol{\alpha}^2$. The $\begin{bmatrix} \boldsymbol{K}_{BB}, \boldsymbol{K}_{BN} \\ \boldsymbol{K}_{BN}^T, \boldsymbol{K}_{NN} \end{bmatrix}$ represents the permutation of $\boldsymbol{K}$ according to the selected working set. The $C(\alpha_i^1)$ indicates the upper bound of α_i^1. Analogous to the working set selection strategy in the literature [40], two variables from $\boldsymbol{\alpha}^1$ or $\boldsymbol{\alpha}^2$ are selected as working sets $\boldsymbol{\alpha}_B^1$ or $\boldsymbol{\alpha}_B^2$ to update the dual variables. Note that $\boldsymbol{\alpha}^1$ and $\boldsymbol{\alpha}^2$ constrained by the same equation. Taking selecting variables from $\boldsymbol{\alpha}^1$ as the example, we should select the variables that present the steepest descent direction, i.e., the variables that can minimize the following optimization problem:

$$\begin{aligned}
&\min_{\boldsymbol{d}} \nabla f(\boldsymbol{\alpha}_1)^T\boldsymbol{d} \\
&\text{s.t. } \sum_i y_i^1 d_i = 0 \\
&\quad d_i \ne -1, \quad \text{if } \alpha_i^1 = 0 \\
&\quad d_i \ne 1, \quad \text{if } \alpha_i^1 = C\left(\alpha_i^1\right) \\
&\quad \#\{d_i | d_i \ne 0, 1 \le i \le N_1\} = 2 \\
&\quad d_i \in \{1, 0, -1\} 1 \le i \le N_1.
\end{aligned} \tag{28}$$

The non-zero elements of variable $\boldsymbol{d}$ denote the updating direction of working variables. Observing (28), it can be found that the feasible solution of $\boldsymbol{d}$ can be expressed as $\{\boldsymbol{d} | d_{i_1} = 1, d_{i_2} = -1, y_{i_1}^1 = y_{i_2}^1, \alpha_{i_1}^1 < C(\alpha_{i_1}^1), \alpha_{i_2}^1 > 0\}$ and $\{\boldsymbol{d} | d_{i_1} = -1, d_{i_2} = -1, y_{i_1}^1 = -y_{i_2}^1, \alpha_{i_1}^1 > 0, \alpha_{i_2}^1 > 0\}$, and $\{\boldsymbol{d} | d_{i_1} = 1, d_{i_2} = 1, y_{i_1}^1 = -y_{i_2}^1, \alpha_{i_1}^1 < C(\alpha_{i_1}^1), \alpha_{i_2}^1 < C(\alpha_{i_2}^1)\}$. To obtain a fast descent, the (i_1, i_2) corresponds to the non-zero entries of feasible $\boldsymbol{d}$ that minimize the objective function is chosen as the working set. By ignoring the bound constraint, embed the equation constraint implicitly into the objective function as:

$$\begin{aligned}
\min_{d_{\text{step}}} f\left(d_{\text{step}}\right) = &\left(\boldsymbol{\alpha} + d_{\text{step}} \times \boldsymbol{d}\right)^T \boldsymbol{K}\left(\boldsymbol{\alpha} + d_{\text{step}} \times \boldsymbol{d}\right) \\
&- \boldsymbol{e}^T\left(\boldsymbol{\alpha} + d_{\text{step}} \times \boldsymbol{d}\right).
\end{aligned} \tag{29}$$

Let the derivatives of Eq. (29) be zeros. We have

$$d_{\text{step}}^* = \frac{-2\boldsymbol{d}^T\boldsymbol{K}\boldsymbol{\alpha} + \boldsymbol{e}^T\boldsymbol{d}}{2\boldsymbol{d}^T\boldsymbol{K}\boldsymbol{d}}. \tag{30}$$

Then, combining with bound constraints, select d_{step} that is nearest to d_{step}^* as the final updating step size

$$d_{\text{step}} = \begin{cases} \min\left(d_{\text{step}}^*, C\left(\alpha_{i_1}^1\right) - \alpha_{i_1}^1, C\left(\alpha_{i_2}^1\right) - \alpha_{i_2}^1\right), \\ \quad \text{if } \left[d_{i_1}, d_{i_2}\right] = [1, 1] \\ \min\left(d_{\text{step}}^*, \alpha_{i_2}^1, \alpha_{i_1}^1\right), \\ \quad \text{if } \left[d_{i_1}, d_{i_2}\right] = [-1, -1] \\ \min\left(d_{\text{step}}^*, C\left(\alpha_{i_1}^1\right) - \alpha_{i_1}^1, \alpha_{i_2}^1\right), \\ \quad \text{if } \left[d_{i_1}, d_{i_2}\right] = [1, -1]. \end{cases} \tag{31}$$

In accordance with the above equation, update dual variables:

$$\boldsymbol{\alpha}_1 = \boldsymbol{\alpha}_1 + d_{\text{step}} \times \boldsymbol{d}. \tag{32}$$

Repeat the selection of the next working set until the following terminal criteria are met:

$$\nabla f(\boldsymbol{\alpha}_1)^T\boldsymbol{d} \le \delta_d \tag{33}$$

where δ_d denotes the pre-established manually positive tolerance. When the $\nabla f(\boldsymbol{\alpha}_1)^T\boldsymbol{d}$ of current working set is smaller than the δ_d, stop iteration and output the optimal $\boldsymbol{\alpha}$.

IV. Online Domain Adaptation Support Tensor Machine

A. Online Version of DA-STM

When the set of samples from a specific source is too small, the DA-STM is difficult to be trained effectively. Consider the case of multisource satellite sequential observations. To make full use of the sequentially acquired observation samples, if a source lacks sufficient samples, online samples from the corresponding source should be fed into the DA-STM to update the model parameters to improve cross-source classification performance. To achieve this intention, the DA-STM is upgraded to the online DA-STM (ODA-STM) in this section.

For the sequence observation case, the researchers have constructed different online SVM methods to update model parameters online. For example, non-convex online SVM was built combined with slope loss to improve the robustness of outlier, resulting in better sequence recognition results [41]. To improve recognition performance with the acquisition of sequence sample, the online STM (OSTM) was established using stochastic gradient method to update model online [42].

Compared with the existing online SVM and STM methods that only need to update the hyperplane parameters of single-source data, DA-STM needs to update both the shared hyperplane parameters and factor matrices using cross-source heterogeneous data. Given the samples $\mathcal{X}_k^t$ from source k ($k \in \{1, 2\}$) acquired in the tth round, to improve the DA-STM performance gradually, it is necessary to refine the DA-STM parameters as the acquisition of sequential samples. Therefore, by integrating the passive-aggressive (PA) strategy [43], the online DA-STM (ODA-STM) is proposed to update the model parameters according to the sample acquired in the tth round, and the corresponding optimization problem is:

$$\min_{\mathcal{G}, \boldsymbol{W}_1^l, \boldsymbol{W}_2^l, b_1, b_2} \frac{1}{2}\left\| \mathcal{G}\prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l - \widehat{\mathcal{G}}\prod_{l=1}^{L} \times_l \widehat{\boldsymbol{W}}_1^l \right\|_F^2$$

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$$
\begin{aligned}
&+ \frac{1}{2}\left\|\mathcal{G}\prod_{l=1}^{L}\times_l \boldsymbol{W}_2^l - \widehat{\mathcal{G}}\prod_{l=1}^{L}\times_l \widehat{\boldsymbol{W}}_2^l\right\|_F^2 \\
&+ \frac{1}{2}\left\|[b_1, b_2] - \left[\widehat{b}_1, \widehat{b}_2\right]\right\|_F^2 \\
&+ C_1\xi_1 + C_2\xi_2 \\
\text{s.t. } & y_k^t\left(\left\langle \mathcal{G}\prod_{l=1}^{L}\times_l \boldsymbol{W}_k^l, \mathcal{X}_k^t\right\rangle + b_k\right) \geq 1 - \xi_k \\
& \xi_k \geq 0, \quad k \in \{1, 2\}
\end{aligned} \tag{34}
$$

where $\widehat{\mathcal{G}}$, $\widehat{\boldsymbol{W}}_1^l$, $\widehat{\boldsymbol{W}}_2^l$, and $[\widehat{b}_1, \widehat{b}_2]$ denote the shared core tensor updated in the last round, the factor matrix of source 1 updated in the last round, the factor matrix of source 2 updated in the last round, and the bias updated in the last round, respectively. For the optimization problem of ODA-STM, the objective is to make the updated hyperplane parameters effectively separate the currently acquired sequence samples by maintaining the original hyperplane parameters as much as possible.

Noting that Eq. (34) is non-convex optimization, the alternating optimization technique is used to optimize the shared core tensor and the factor matrix sequentially. The solving procedure of ODA-STM is similar to that of DA-STM, and the detailed derivations can be found in the Appendix.

B. Analysis of Performance for DA-STM and ODA-STM

Compared to DA-STM, ODA-STM can continuously update the model parameters using multisource sequence samples to obtain better classification performance. To prove this ability, the performance of DA-STM and ODA-STM is analyzed elaborately in this section.

For the ODA-STM, the loss in round t is defined by

$$
l_t\left(\mathcal{G}, \boldsymbol{W}_1^l, \boldsymbol{W}_2^l, b_1, b_2\right) = C_1\xi_1 + C_2\xi_2. \tag{35}
$$

To express that ODA-STM may obtain better classification results compared to DA-STM, the following two theorems are constructed.

Theorem 1: For the case of sequence sample acquisition, use the DA-STM with parameters $H_0 = \{\mathcal{G}, \boldsymbol{W}_1^l, \boldsymbol{W}_2^l, b_1, b_2\}$ to initialize the ODA-STM. When acquiring a sample in round t, the model parameters of ODA-STM are updated to $H_t = \{\mathcal{G}_t, \mathcal{W}_1^t, \mathcal{W}_2^t, b_1^t, b_2^t\}$. Then, the loss in round t of ODA-STM has the following relationship compared to the loss of DA-STM:

$$
l_t(H_0) - l_t(H_t) \geq \sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^0\right\|_F^2 + \sum_{m=1}^{2}\left\|b_m^t - b_m^0\right\|_F^2. \tag{36}
$$

Proof: For the ODA-STM initialized with H_0, there exist two cases. In case 1, if $l_t(H_0) = 0$, it is no need to update parameters, thus we have $l_t(H_t) = 0$. At this point, theorem 1 holds. In case 2, if $l_t(H_0) > 0$, according to the optimization problem of ODA-STM, it is known that the value of the objective function with H_0 is larger than that with parameter H_t, that is:

$$
\begin{aligned}
& l_t(H_t) + \sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^0\right\|_F^2 + \sum_{m=1}^{2}\left\|b_m^t - b_m^0\right\|_F^2 \\
& \leq l_t(H_0) + \sum_{m=1}^{2}\left\|\mathcal{W}_m^0 - \mathcal{W}_m^0\right\|_F^2 + \sum_{m=1}^{2}\left\|b_m^0 - b_m^0\right\|_F^2.
\end{aligned} \tag{37}
$$

Therefore, Theorem 1 holds obviously. Theorem 1 indicates that updating the model parameters using ODA-STM for the current sample results in a smaller loss compared to DA-STM, i.e., a better recognition ability. Subsequently, the following theorem is constructed to estimate the upper bound of loss for the ODA-STM.

Theorem 2: When using ODA-STM for sequence updating, assume that there exist model parameters $H_* = \{\mathcal{W}_1^*, \mathcal{W}_2^*, b_1^*, b_2^*\}$ such that their loss $l_t(H_*) = 0$ for all $1 \leq t \leq T$. Then, we have:

$$
\sum_{t=1}^{T} l_t(H_t) \leq \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^*\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^t - b_m^*\right\|_F^2. \tag{38}
$$

Proof: Since H_t are the optimal parameters of ODA-STM, and $l_t(H_*) = 0$ for all $1 \leq t \leq T$, we have:

$$
\begin{aligned}
&\sum_{t=1}^{T} l_t(H_t) + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^{t-1}\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^t - b_m^{t-1}\right\|_F^2 \\
&\leq \sum_{t=1}^{T} l_t(H_*) + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^* - \mathcal{W}_m^{t-1}\right\|_F^2 \\
&\quad + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^* - b_m^{t-1}\right\|_F^2
\end{aligned} \tag{39}
$$

$$
\begin{aligned}
&\sum_{t=1}^{T} l_t(H_t) + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^{t-1}\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^t - b_m^{t-1}\right\|_F^2 \\
&\quad - \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^* - \mathcal{W}_m^{t-1}\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^* - b_m^{t-1}\right\|_F^2 \leq 0.
\end{aligned} \tag{40}
$$

According to the trianglar inequality property of norm, we have:

$$
\begin{aligned}
&\sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^{t-1}\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^t - b_m^{t-1}\right\|_F^2 \\
&\quad - \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^* - \mathcal{W}_m^{t-1}\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^* - b_m^{t-1}\right\|_F^2 \\
&\leq \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|\mathcal{W}_m^t - \mathcal{W}_m^*\right\|_F^2 + \sum_{t=1}^{T}\sum_{m=1}^{2}\left\|b_m^t - b_m^*\right\|_F^2.
\end{aligned} \tag{41}
$$

Therefore, Theorem 2 holds.

Theorem 2 gives the upper bound of loss for ODA-STM under certain conditions. According to the above theorems, it can be found that ODA-STM can obtain better cross-source recognition results compared with DA-STM for multisource sequence observation case.

Subsequently, the computational complexity of DA-STM is analyzed. Given N L-order cross-source tensor samples, the training DA-STM can be divided two stages. In the first stage, the kernel matrix $K \in \mathbb{R}^{N\times N}$ needs to be constructed, where each entry of $\boldsymbol{K}$ is the inner product between projected tensor

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samples. Therefore, the computational complexity of calculating $\boldsymbol{K}$ is $o(N^2 I^L)$, where I denotes the largest dimension of tensor sample. In the second stage, the dual problem of DA-STM needs to be solved. For the proposed decomposition algorithm, the main computational consumption is used to select the proper working set from $N' \times N'$ variable pair, where $N' \leq N$ denotes the number of dual variables that satisfies the constraints of (28). Therefore, the time complexity for the second stage of training DA-STM is $o(N' \times N' t)$, where t denotes the number of iterations of the decomposition algorithm. Therefore, the total time complexity of training DA-STM is $o(k_{\text{th}}(N^2 I^L + N' \times N' t))$, where k_{th} denotes the maximum iteration number of training different modes of DA-STM.

For the ODA-STM, the main time consumption is to calculate the dual variable α to update model parameter. When calculate α, it is necessary to calculate the projected tensor sample $\widetilde{\boldsymbol{X}}_t$ and variable $\widetilde{\boldsymbol{A}}$, whose time complexity is probably $o(2I^L)$. Assume the ODA-STM is trained by T times. The total computational complexity of ODA-STM is $o(2I^L T)$.

The relationship between the proposed DA-STM, ODA-STM, and the previous methods can be summarized as follows. ODA-STM can be considered as the online extension of DA-STM, and DA-STM belongs to the cross-source domain adaptation classifier. When only a single source exists, the DA-STM degenerates to the support Tucker machine by ignoring the shared core tensor and adaptive class labels [44]. In addition, when the projection tensor is further simplified as a rank-1 tensor, the DA-STM degenerates to the standard C-STM. When the projection tensor is represented as a vector pattern, the proposed method degenerates to the standard C-SVM. For the shared core tensor design of the projection tensor, which is similar to our previous work on coupled heterogeneous Tucker decomposition (CHTD) [45], the method can be used for feature-level transfer learning.

V. Experiments and Analysis

To fully validate the performance of the proposed DA-STM method, we construct four datasets and evaluate the effectiveness of DA-STM compared with typical domain adaptation SVM and STM methods for cross-domain object recognition. The detailed information on these three datasets is as follows.

1) Dataset 1 is composed of optical remote sensing images with different resolutions collected from Google Earth service, which can be divided into source 1 subset and source 2 subset. The source 1 subset contains 168 ship slices of four types with a spatial resolution of 0.5 m, of which the number of ships of types 1, 2, 3, and 4 is 35, 39, 50, and 44, respectively. The source 2 subset contains 168 ship slices of four types with a spatial resolution of 1 m, of which the number of ships of types 1, 2, 3, and 4 is 35, 39, 50, and 44, respectively. These ship slices with different spatial resolutions are cut manually from different harbor regions. They present different backgrounds and thus have different feature distributions. The representative slices of dataset 1 are displayed in Fig. 2.

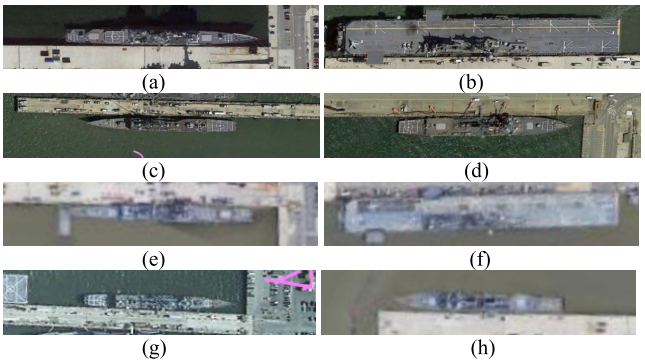


Fig. 2. Representative ship slices with different spatial resolutions. (a)–(d) Display the ship slices of four types with 0.5-m resolution, and (e)–(h) display the ship slices of four types with 1-m resolution.

2) Dataset 2 contains optical airplane slices observed from two types of satellites, i.e., Superview-1 satellite and Jilin-1 satellite. These two types of satellite data present different spatial resolutions and imaging angles. Specifically, the source 1 data (i.e., Superview-1 satellite data) contains 36 airplane slices of three types with 0.5-m resolution and roll satellite angle less than 20°, of which the number of airplanes of types 1, 2, and 3 is 20, 5, and 11, respectively. The source 2 data (i.e., Jilin-1 satellite data) contains 65 airplane slices of three types with 1-m spatial resolution and roll satellite angle greater than 20°, of which the number of airplanes of types 1, 2, and 3 is 36, 5, and 24, respectively. These object slices are cut manually from two groups of satellite images. The specific information of the used Superview-1 and Jilin-1 satellite images is shown in Table I. The representative slices observed from different satellites are given in Fig. 3.
3) Dataset 3 is the publicly available moving and stationary target acquisition and recognition (MSTAR) dataset [46], which contains SAR vehicle slices with a depression angle of 17° and a spatial resolution of 0.3 m × 0.3 m from X-band. Since the MSTAR dataset contains SAR slices with different aspect angles, we selected SAR slices with a significant difference in aspect angles to build source 1 subset, source 2 subset, and source 3 subset for cross-source object recognition. The source 1 sub-set contains 177 SAR slices with aspect angles ranging from 0° to 45°, the source 2 sub-set contains 177 SAR slices with aspect angles ranging from 120° to 165°, and the source 3 sub-set contains 177 SAR slices with aspect angles ranging from 240° to 285°. Both the source 1, 2, and 3 sub-sets contain five types of vehicle targets, namely 2S1-B01, BRDM2, T62, ZIL131, and T72. The number of these vehicles of each type is 37, 37, 37, 37, and 29, respectively. The representative vehicles are shown in Fig. 4.
4) Dataset 4 is the cross-scene EShanghai-EHangzhou HSIs dataset [47]. These two HSIs are acquired from EO-1 Hyperion hyperspectral sensor. In detail, the size of EO-1 Shanghai (EShanghai) and EO-1 Hangzhou (EHangzhou) HSI is 1600 × 230 × 198 and 590 × 230 × 198, respectively. Both HSIs contain three

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TABLE I
DETAILS OF MULTISOURCE REMOTE SENSING IMAGES OBSERVED FROM SUPERVIEW-1 SATELLITES AND JILIN-1 SATELLITE

satellite	roll angle	resolution	acquired time
SuperView-1	-2.79 degree	0.5 meter	02.09.2020
SuperView-1	-13.62 degree	0.5 meter	07.30.2020
Jilin-1	33.11 degree	1 meter	10.07.2020
Jilin-1	-34.20 degree	1 meter	11.07.2020

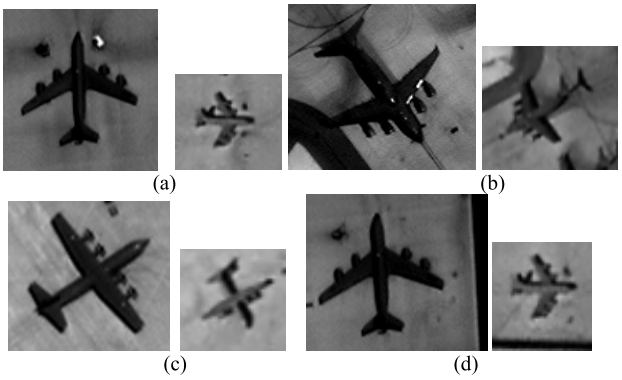


Fig. 3. Representative multi-angle and multi-resolution airplane slices obtained from Superview-1 and Jilin 1 satellites data in dataset 2. In (a)–(d), the left airplane slice is obtained from Superview-1 satellite and the right airplane slice is obtained from Jilin 1 satellites.

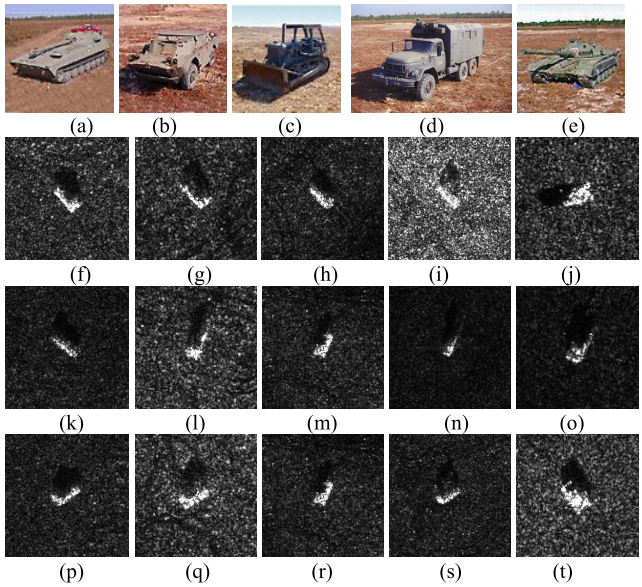


Fig. 4. Representative multi-angle SAR samples obtained from dataset 3. (a)–(e) display the optical vehicle slices of five types. (f)–(j) display SAR vehicle slices with aspect angle 0° to 45° from source 1. (k)–(o) display SAR vehicle slices with aspect angle 120° to 165° from source 2. (p)–(t) display SAR vehicle slices with aspect angle 240° to 285° from source 3.

land-cover classes, i.e., Water, Grounding, and Plant. The data cubes of cross-scene HSIs and their ground truth maps can be found in Fig. 5.

Experiments are conducted in four sections. In Section V-A, the feature tensor construction method is described to obtain the feature representation of cross-source data. In Section V-B, the impact of parameter setting on the proposed DA-STM is analyzed, and the effectiveness of the online update strategy of ODA-STM is evaluated. In Section V-C, the time consumption of the proposed decomposition algorithm is examined and compared with two typical optimization solvers.

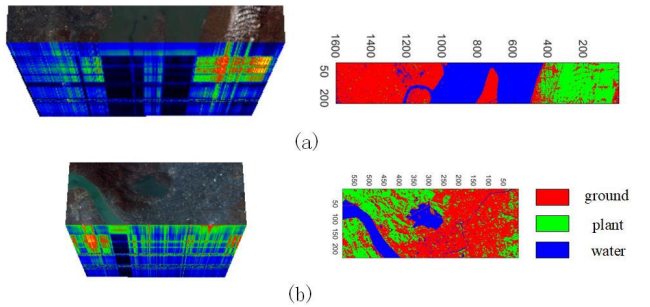


Fig. 5. Cross-scene HSIs cubes and the corresponding groundtruth maps. (a) denotes the EShanghai HSI and its groundtruth and (b) denotes the EHangzhou HIS and its groundtruth.

In Section V-D, the superiority of the proposed DA-STM and ODA-STM for cross-source object recognition is verified compared with typical domain adaptation SVMs and STMs. In Section V-E, the performance of the proposed DA-STM is compared with three state-of-the-art deep learning methods. All simulations are executed on MATLAB 2018 software on a computer with a Windows 10 operating system equipped with a CPU of i5-7300HQ processor and 8 GB memory.

A. Construction of Multisource Feature Tensors

To adequately characterize the discriminative features of object slices from different source, multiple feature extraction methods are utilized separately to extract texture, shape, and scattering features of images, and then these features are reshaped to feature tensor with three orders, i.e., the horizontal spatial order, vertical spatial order, and feature order. The detailed feature extraction and feature tensor construction methods are described as follows.

1) Gabor Feature Tensor [48]: Gabor features record the filter responses at different scales and orientations, which can accurately capture the local spatial properties of the object. In the experiments, the Gabor filters with different scales $\{0, 1, 2, 3, 4\}$ and orientations $\{0, 1, 2, 3, 4\}$ are used to obtain the energy responses of the input slices through convolution operations, resulting in a third-order feature tensor $\mathcal{F}_{\text{gabor}} \in \mathbb{R}^{W\times H\times 40}$, where the first and second orders are spatial orders, and the third order is the feature order.

2) Local Binary Pattern (LBP) Feature Tensor [49]: LBP features describe the relative differences between pixels at different locations and their neighbors. In the experiments, LBP features with different radii $\{1, 2, 3\}$ are extracted and form the three-order tensor $\mathcal{F}_{\text{LBP}} \in \mathbb{R}^{W\times H\times 3}$, where the first and second orders are spatial orders, and the third order is the feature order.

3) Morphological Feature Tensor [50]: The morphological top-hat and bottom-hat operators can highlight the pixels with local gray extremes in the slice, effectively describing the local shape characteristics of the object. In the experiments, by using the morphological top-hat and bottom-hat operators, the morphological feature tensor $\mathcal{F}_{\text{mor}} \in \mathbb{R}^{W\times H\times 2}$ is constructed, where the first order, the second order, and the third order denote the horizontal spatial order, the vertical spatial order, and the feature order, respectively.

4) Scattering Feature Tensor [51]: The intensity of the backward scattering characteristics of SAR images reflects

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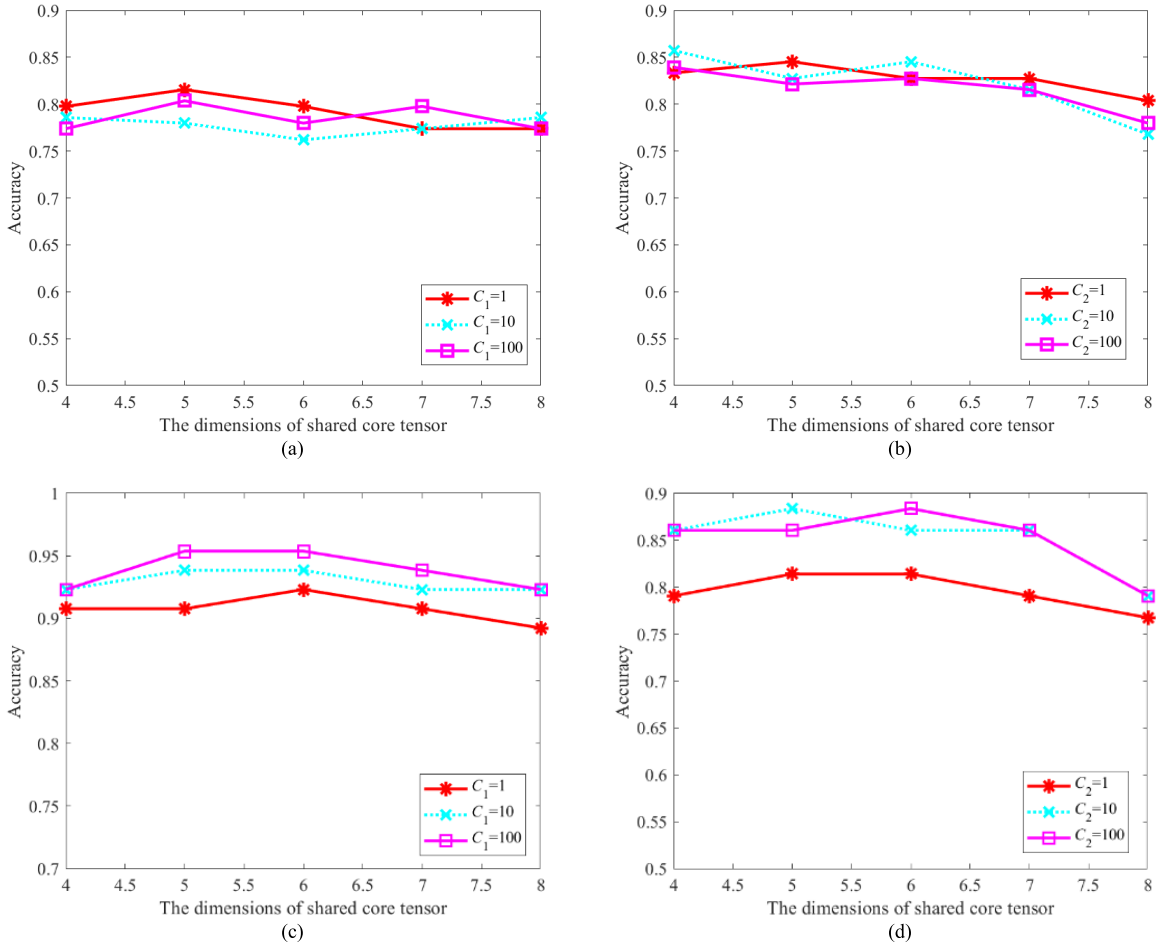


Fig. 6. Recognition results of DA-STM under different parameter settings. (a) displays the recognition accuracy of samples from source 1 of dataset 1 using DA-STM with $C_2 = 1$. (b) gives the recognition accuracy of samples from source 2 of dataset 1 with $C_1 = 1$. (c) displays the recognition accuracy of samples from source 1 of dataset 2 using DA-STM with $C_2 = 1$. (d) gives the recognition accuracy of samples from source 2 of dataset 2 with $C_1 = 1$.

the geometry and materials of the object. In the experiments, the gray-level co-occurrence matrices (GLCM) features of different orientations $\{[2, 0], [0, 2], [1, 1], [1, -1]\}$ are extracted, and the third-order scattering feature tensor $\mathcal{F}_{\text{GLCM}} \in \mathbb{R}^{W\times H\times 4}$ is constructed, where the first order, the second order, and the third order denote the horizontal spatial order, the vertical spatial order, and the feature order, respectively.

It is known that data from different sources should be characterized by different feature tensors. To obtain the best object recognition results, multisource feature tensors are built separately [52]. For optical object slices, all feature tensors except the scattering feature tensor are cascaded along with the third order as the optical feature tensor. For SAR object slices, the scattering feature tensor is used as the SAR feature tensor.

Unlike the first three datasets, it is worth noting that the HSIs in dataset 4 can be naturally represented as three-order tensors. In the experiments, the small cubes sized of $5 \times 5 \times 198$ centered on the different pixels of HSIs is extracted as the feature tensors to be identified, and the label is determined by the land-cover class of the center pixel.

B. Analysis of the Impact of Parameter Setting on DA-STM

The parameter of DA-STM includes the regularization parameter C_1 of source 1, the regularization parameter C_2 of source 2, the regularization parameter C_3 of unlabeled samples, and the dimension of the shared core tensor $\{i_l | 1 \le l \le L\}$. To analyze the impact of different parameters on DA-STM, the experiments are conducted using the N-fold cross-validation method for performance evaluation. This involves dividing the sample set into N portions, where $N - 1$ portions are used as the training set, and the remaining one is used as the test set. The training and test sets are then alternated until each sample subset has been used as a test set once. First, by adjusting different parameter settings, the experiments verify the recognition performance of DA-STM under different C_1, C_2, and the dimension of shared core tensor using dataset 1 and dataset 2, as shown in Fig. 6.

In Fig. 6, the horizontal axis j denotes the dimension of the shared core tensor $i_l = \text{ceil}((1/j) \times I_l)$. As seen in Fig. 6, there is a relationship between the recognition results of DA-STM and different regularization parameters. For data from dataset 1, the impact of C_1 and C_2 on the recognition results is unobvious. For data from dataset 2, it is seen that the recognition results improve as the regularization parameter C_1 increases, and the recognition results improve as the regularization parameter C_2 increases. This finding indicates that the regularization parameter should be set according to the specific dataset. In addition, for both the two datasets, it can be found that the recognition accuracy decreases as

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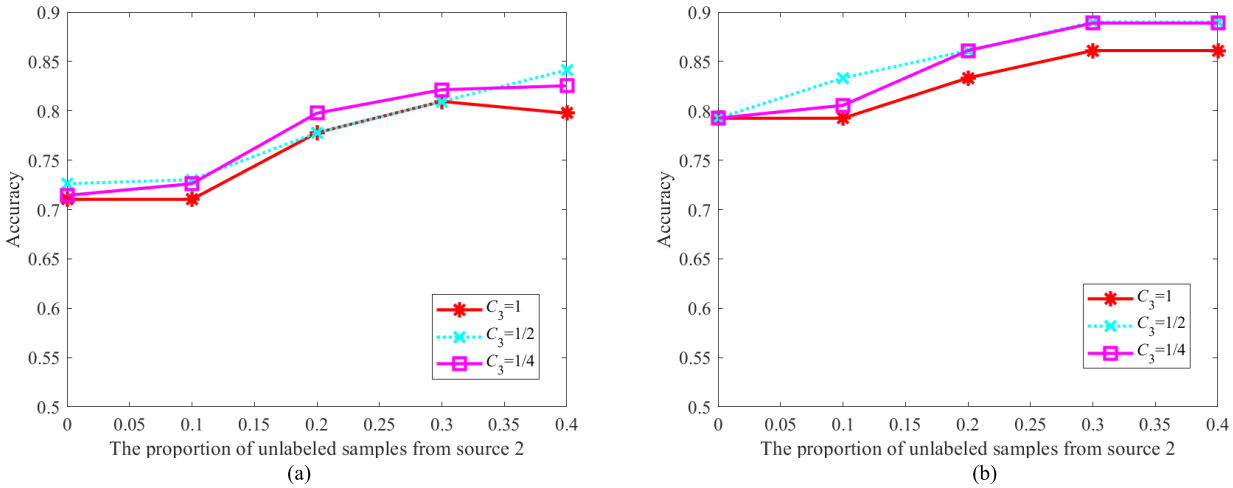


Fig. 7. Accuracy of DA-STM under different proportions of unlabeled samples. (a) denotes the results of different C_3 for DA-STM using dataset 1. (b) denotes the results of different C_3 for DA-STM using dataset 2.

the dimension of the shared core tensor decreases because it is difficult to generate an effective classification hyperplane with a small-size shared core tensor to accurately separate objects from different sources. Therefore, it is necessary to use a shared core tensor of an appropriate size to ensure more effective recognition results.

The DA-STM can absorb useful information from unlabeled samples to improve cross-source target recognition results. To verify this ability, source 2 subset of dataset 1 (dataset 2) is split into two equal parts, one for labeled samples and one for unlabeled samples. Then, set different values of C_3, and add different numbers of unlabeled samples to the training set for the training of DA-STM by maintaining C_1 and C_2 be equal to 1. The resulting recognition accuracy is shown in Fig. 7.

In Fig. 7, the horizontal axis indicates the proportion occupied by unlabeled samples. It can be seen that the accuracy of DA-STM improves with the addition of unlabeled samples. It indicates that DA-STM can explore potential category information from the unlabeled samples to improve the quality of the learned classification hyperplane. In addition, by comparing different C_3, it can be found that the best results of DA-STM with $C_3 < 1$ because the larger C_3 may bring samples with false labels to adversely affect its recognition performance, which indicates that the weights of unlabeled samples should not be overly large to ensure the effectiveness of the classification hyperplane.

C. Evaluation of the Time Consumption of the Proposed Decomposition Algorithm Compared With Typical Optimization Solvers

To accelerate the training of DA-STM, analogous to SMO for SVM, the DA-STM-oriented decomposition algorithm is proposed to break the original complex quadratic programming into a series of simple analytic optimization problems, thus accelerating the training of the model. To verify the effectiveness of the proposed decomposition algorithm for model training, the experiments are carried out on dataset 2 to evaluate the time consumption of the proposed decomposition algorithm, the interior-point method [53], and the active set method [54]. In detail, the training set is constructed using different ratios of dataset 2 samples, and the maximum number of iterations of DA-STM is set to 10. After training, the rest samples are used as test samples to examine the recognition accuracy of DA-STM trained by different solving methods. The obtained results are displayed in Fig. 8 and Table II.

TABLE II

ACCURACIES OF DA-STM USING DIFFERENT OPTIMIZATION SOLVING METHODS FOR DIFFERENT RATIOS OF TRAINING SET

	50%	60%	70%	80%	90%
interior-point method	83.33%	84.61%	89.47%	84.61%	85.71%
active-set method	83.33%	84.61%	89.47%	84.61%	85.71%
proposed	83.33%	84.61%	89.47%	84.61%	85.71%

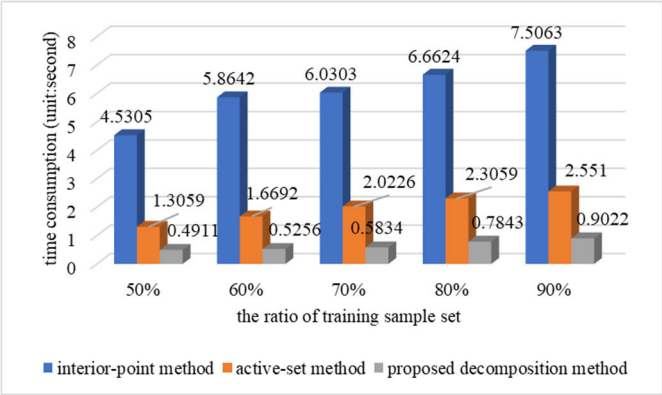


Fig. 8. Time consumption of the proposed decomposition algorithm compared with typical optimization solvers.

According to Fig. 8, it can be found that the proposed decomposition algorithm reduces time consumption by more than 50% compared with the typical two optimization solvers for different ratios of training sets. This is due to the fact that the proposed decomposition algorithm replaces the complex numerical optimization of the traditional optimization solvers with a series of simple analytic optimizations. Specifically, the time consumption is positively correlated with the training size for all the solving methods because a more significant number of training samples implies a larger search space for the dual problem. In addition, the dual problem of the proposed DA-STM is convex. That means different optimization solving

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methods can converge to the same optimal solution. Thus, it is seen from Table II that these methods get the same recognition accuracy. Remarkably, the proposed decomposition algorithm maintains a low time consumption under different ratios of training set and gets the same recognition accuracy as the competitors, ensuring the efficient implementation of DA-STM.

D. Evaluation of the Performance of the DA-STM Compared With Typical SVM and STM Classifiers

To comprehensively evaluate the performance of the proposed DA-STM, experiments are conducted on three datasets to evaluate DA-STM compared with different domain adaption SVM and STM for cross-source object recognition. The information on the used comparison methods is introduced as follows.

1) *SVM:* SVM is a classical vector pattern classifier. In the experiments, the SVM is trained directly by using source 1 or 2 data to generate two individual classifiers for object recognition from different sources.
2) *TCA+SVM:* TCA is a classical unsupervised domain adaptation method. In the experiments, the classical linear kernel-based TCA method is used to transfer data from different sources to the same space, and then the SVM is trained using the transferred samples.
3) *ATL+SVM:* ATL can jointly select representative samples and discriminant features for the supervised domain adaptation. In the experiments, ATL is adopted using the default parameters in the public code to transfer the data from different sources to the same feature space. Then the SVM is trained using the transferred samples.
4) *CMMS+SVM:* The centroid matching and local manifold self-learning (CMMS) method uses category center matching with local manifold adaptive embedding to achieve domain adaptation [55]. In the experiments, CMMS is implemented using the default parameters in the public code to transfer the data from different sources to the same feature space first, and then the SVM is trained using the transferred samples.
5) *DJDA+SVM:* DJDA is a heterogeneous domain adaptation method [35]. It utilizes the distribution align and the discriminative least square regression to obtain domain adaption features. Then, the SVM is trained using these features to achieve cross-source recognition.
6) *ASVM:* ASVM is a classical domain adaptation classifier. In the experiments, the PCA is employed to adjust the space of sources 1 and 2 to the same dimension. The samples from source 1 are applied to initialize the auxiliary classifier of ASVM, and then the samples from source 2 are used to train the delta function of ASVM.
7) *DTMKL:* DTMKL is the kernel version of the domain adaptation classifier. In the experiments, the linear kernel, the Gaussian kernel, and the polynomial kernel are applied for the DTMKL. To map the heterogeneous data to the same dimension, the PCA is used to map the samples of source 1 and source 2 to the same space. Then, the DTMKL classifier is trained using the transferred samples.

TABLE III
OA OF DIFFERENT CLASSIFIERS ACROSS SOURCES FOR DATASET 1

method	parameter setting	accuracy (source 1)	parameter setting	accuracy (source 2)
SVM	C=10	58.93%	C=10	60.17%
TCA+SVM	C=10	64.88%	C=1	64.88%
ATL+SVM	C=10	77.98%	C=1	80.36%
CMMS+SVM	C=100	68.45%	C=100	64.88%
DJDA+SVM	C=10	77.38%	C=1	80.36%
ASVM	C_1=10,C_2=10	79.76%	C_1=10,C_2=10	80.36%
DTMKL	C=1	51.19%	C=1	54.76%
STM	C=10	57.74%	C=10	58.33%
CHTD+STM	C=10	71.43%	C=10	72.62%
DA-STM(semi-supervised)	C_1=1,C_2=10, C_3=1	76.19%	C_1=1,C_2=10, C_3=1	82.74%
DA-STM	C_1=100,C_2=1	81.55%	C_1=1,C_2=10	85.72%

8) *STM:* STM is the classical tensor pattern classifier. In the experiments, the samples from sources 1 or 2 are used to train STM separately, generating two recognition models for sources 1 and 2, respectively.
9) *CHTD+STM:* CHTD is a tensor-based domain adaptation method. In the experiments, the CHTD is used to extract the shared feature tensor across sources, and then the STM is trained using the transferred samples to evaluate the cross-source recognition results.

The used evaluation metrics include accuracy per class, overall accuracy (OA), and Kappa coefficients. The OA measures the overall recognition ability of the method for all the test samples, as calculated by (N_r/N), of which N_r and N denote the number of correctly identified objects and the total number of the objects, respectively. Note that datasets 1, 2, 3, and 4 present different class distributions. To measure recognition performance fairly, we use the Kappa coefficient as an additional metric to evaluate different recognition methods. The Kappa coefficient [35], another widely used measure of recognition ability, can better reflect recognition ability regardless of imbalanced class distributions. Using these three evaluation metrics can verify our DA-STM comprehensively.

For the proposed method, the performance of DA-STM and DA-STM (semi-supervised) is examined separately. In DA-STM, all training samples used are labeled. In semi-supervised DA-STM, half of the training samples used are labeled, and the other half are unlabeled. The comparison method includes vector-based domain adaption classifier and tensor-based domain adaption classifier. To obtain a fair comparison, the feature tensor or feature vector is set to the same dimension for all comparison methods. To obtain the best results for different classifiers, adjust the parameter settings for each method, including the regularization parameters $C \in \{1, 10, 100\}$. Using twofold cross-validation, the final recognition results for datasets 1 and 2 are shown in Tables III and IV, respectively.

According to the results in the above tables, it can be found that for the three datasets, direct application of SVM or STM without the domain adaptation method will yield poor results due to the differences in resolution or imaging angle, or aspect angle across sources, which makes it difficult for

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TABLE IV
OA OF DIFFERENT CLASSIFIERS ACROSS SOURCES FOR DATASET 2

method	parameter setting	accuracy (source 1)	parameter setting	accuracy (source 2)
SVM	C=10	67.69%	C=1	52.78%
TCA+SVM	C=1	72.31%	C=10	55.56%
ATL+SVM	C=1	80.00%	C=10	63.89%
CMMS+SVM	C=100	76.92%	C=10	69.44%
DJDA+SVM	C=10	90.77%	C=1	80.56%
ASVM	C_1=10,C_2=10	90.77%	C_1=1,C_2=10	77.78%
DTMKL	C=1	75.38%	C=1	58.33%
STM	C=100	52.31%	C=10	52.78%
CHTD+STM	C=10	75.38%	C=100	66.67%
DA-STM(semi-supervised)	C_1=100,C_2=1 C_3=10	93.85%	C_1=1,C_2=10, C_3=1	80.56%
DA-STM	C_1=1,C_2=1	93.85%	C_1=1,C_2=100	83.33%

the traditional classifier to perform this cross-source recognition task. Among the feature transfer and SVM combined domain adaptation methods, ATL+SVM and DJDA+SVM obtain effective results in most cases. Because ATL can improve the domain adaptation performance by both domain adaptation projecting at the feature level and outliers removing at the sample level, and DJDA has discriminative least square term that may increase the classification margin of SVM. Furthermore, ASVM also gets acceptable results among the compared methods. This is since ASVM, as a supervised domain adaptation classifier, can fully utilize label information across source samples and achieve the comprehensive utilization of cross-source information by minimizing the delta function. In addition, it can be noticed that these methods performed on samples from source 1 for dataset 2 have higher recognition accuracy than those performed on source 2 for dataset 2 because the number of samples of Jilin 1 is significantly higher than that of Superview-1 data, thus allowing the classifier to be more adequately trained. In addition, it can be found that the proposed DA-STM has a better performance than that of DA-STM (semi-supervised) because DA-STM (semi-supervised) may generate partial mislabeling and thus affect the recognition performance. Notably, the proposed DA-STM outperforms all comparison methods due to the ability of DA-STM to both directly handle heterogeneous cross-source tensor data and improve cross-source recognition results using joint distribution and structure adaptation, which illustrates the quality of the proposed DA-STM.

In addition, the Kappa coefficients are calculated to further evaluate the recognition ability of these methods, as shown in Fig. 9. The DA-STM(ss) in Fig. 9 denotes the semi-supervised DA-STM. It is observed that different methods obtain relatively more minor Kappa coefficients than recognition accuracy because the Kappa coefficients can penalize the unbalanced class distribution, thus evaluating the recognition ability more accurately. Remarkably, it is seen that DA-STM obtains the largest Kappa coefficients among all the methods, further demonstrating the performance of the DA-STM.

Subsequently, dataset 3 is used to examine further the quality of these classifiers. Note that dataset 3 contains subsets of three sources. To deal with this case, three subsets are

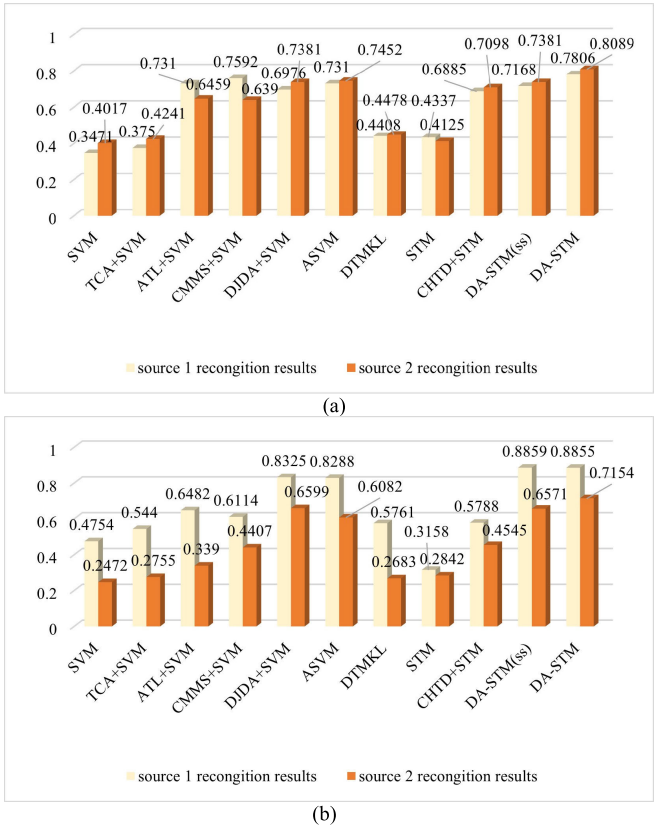


Fig. 9. Kappa coefficients of different domain adaption object recognition methods using different datasets. (a) shows the Kappa coefficients of different methods using dataset 1. (b) shows the Kappa coefficients of different methods using dataset 2.

fed into DA-STM to learn the shared core tensor. For the competitors, they are transferring methods designed for two sources. When recognizing the samples from source 1, the samples from both sources 2 and 3 are treated as a unity to simplify as domain adaptation problem of two sources, and then using the comparison methods to extract cross-source features. Then, using twofold cross-validation, the obtained results are shown in Table V.

From Table V, it is observed that the existing domain adaptation classifiers and the proposed DA-STM can also adapt to the multisource domain adaptation case. Among the competitors, ATL+SVM, CMMS+SVM, DJDA+SVM, and ASVM still get relatively effective results. Since there exists a correlation between different sources, the domain adaption recognition results are better than using SVM or STM directly. Note that our DA-STM also obtains the best results among all the classifiers. It indicates the quality of the DA-STM for multisource domain adaptation.

Considering the HSI can be represented as a tensor naturally, the experiments are further conducted on dataset 4 to examine the performance of DA-STM. In detail, randomly select 30 samples per class from HSI with different scenarios as the training set and 40 samples per class from HIS with different scenarios as the test set. To ensure the reliability of the experiment, the experiment was repeated ten times, and average results were calculated. The confusion matrix obtained is shown in Fig. 10, and the accuracy per class, the OA, and the Kappa coefficient are shown in Tables VI and VII.

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TABLE V

ACCURACIES AND KAPPA COEFFICIENTS OF DIFFERENT CLASSIFIERS FOR DIFFERENT SOURCES USING DATASET 3

	SVM	TCA+SVM	ATL+SVM	CMMS +SVM	DJDA +SVM	ASVM	DTMKL	STM	CHTD+STM	DA-STM (semi-supervised)	DA-STM
OA (source 1)	45.20%	51.41%	80.79%	78.53%	76.27%	77.4%	56.5%	55.37%	77.4%	77.97%	80.79%
Kappa	0.3071	0.3841	0.7598	0.7351	0.7040	0.7178	0.4556	0.4415	0.7177	0.7355	0.7713
OA (source 2)	51.98%	50.85%	71.75%	0.7345%	75.71%	76.27%	57.63%	57.06%	78.53%	79.1%	81.92%
Kappa	0.3962	0.39	0.6467	0.6679	0.6969	0.7037	0.4698	0.4627	0.7313	0.7498	0.7856
OA (source 3)	45.76%	53.76%	0.7627	70.62%	75.71%	75.71%	64.41%	54.8%	78.53%	79.66%	79.66%
Kappa	0.3143	0.4175	0.7033	0.6332	0.6961	0.6967	0.5549	0.4344	0.7317	0.7570	0.7423

TABLE VI

RESULTS OF DIFFERENT INDICATES FOR SOURCE 1 WITH DIFFERENT CLASSIFIERS USING DATASET 4

	SVM	TCA+SVM	ATL+SVM	CMMS+SVM	DJDA +SVM	ASVM	DTMKL	STM	CHTD+STM	DA-STM (semi-supervised)	DA-STM
accuracy (class 1)	90%	95%	70%	72.5%	95%	90%	75%	65%	95%	90%	92.5%
accuracy (class 2)	72.5%	82.5%	92.5%	92.5%	85%	90%	65%	60%	60%	92.5%	92.5%
accuracy (class 3)	42.5%	57.5%	57.5%	45%	77.5%	90%	10%	55%	100%	90%	90%
OA	68.3%	78.3%	73.3%	70%	85.83%	90%	50%	60%	85%	90.83%	91.67%
Kappa	0.5250	0.6750	0.6	0.5272	0.7875	0.85	0.25	0.4	0.7750	0.8625	0.875

TABLE VII

RESULTS OF DIFFERENT INDICATES FOR SOURCE 2 WITH DIFFERENT CLASSIFIERS USING DATASET 4

	SVM	TCA+SVM	ATL+SVM	CMMS+SVM	DJDA +SVM	ASVM	DTMKL	STM	CHTD+STM	DA-STM (semi-supervised)	DA-STM
accuracy (class 1)	92.5%	95%	97.%	75%	95%	97.5%	82.5%	67.5%	87.5%	95%	97.5%
accuracy (class 2)	17.5%	50%	52.5%	97.5%	82.5%	40%	65%	62.5%	92.5%	97.5%	97.5%
accuracy (class 3)	100%	92.5%	100%	42.5%	77.5%	100%	10%	55%	40%	95%	95%
OA	70%	79.17%	83.3%	71.67%	85%	79.17%	52.5%	61.67%	73.3%	95.83%	96.67%
Kappa	0.55	0.6875	0.75	0.5750	0.775	0.6875	0.2875	0.425	0.6	0.9375	0.95

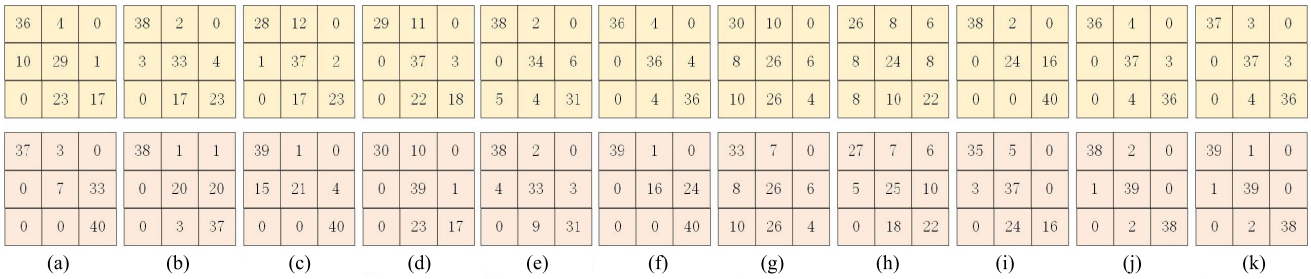


Fig. 10. Confusion matrices of different methods for classification using dataset 4. The upper row displays the results of different methods for classification of source 1, and the lower row displays the results of different methods for classification of source 2. For the confusion matrix, the entry in (i, j) denotes the number of samples of class i that are predicted as class j. (a) SVM. (b) TCA+SVM. (c) ATL+SVM. (d) CMMS+SVM. (e) DJDA+SVM. (f) ASVM. (g) DTMKL. (h) STM. (i) CHTD+STM. (j) DA-STM (semi-supervised). (k) DA-STM.

Observing the results of dataset 4, since the discrepancy between cross-scene HSIs, the direct application of SVM cannot obtain effective results. When using domain adaptation classifiers, the classification results of HSIs are improved. It is seen that the ATL+SVM works well among the competitors. The reason is that the tensor samples of HSIs may contain some mixed pixels, and the ATL can eliminate these mixed samples and improve the transfer results. The ASVM, its performance is still good for dataset 4 because it can incorporate the knowledge from different sources into the hyperplane. The CHTD-STM method also acquires acceptable results because the tensor-based transfer method can exploit the structure information of the HSI patch. Note that our DA-STM has obtained the best results for different indicators using HSI dataset. It further demonstrates the quality of the DA-STM.

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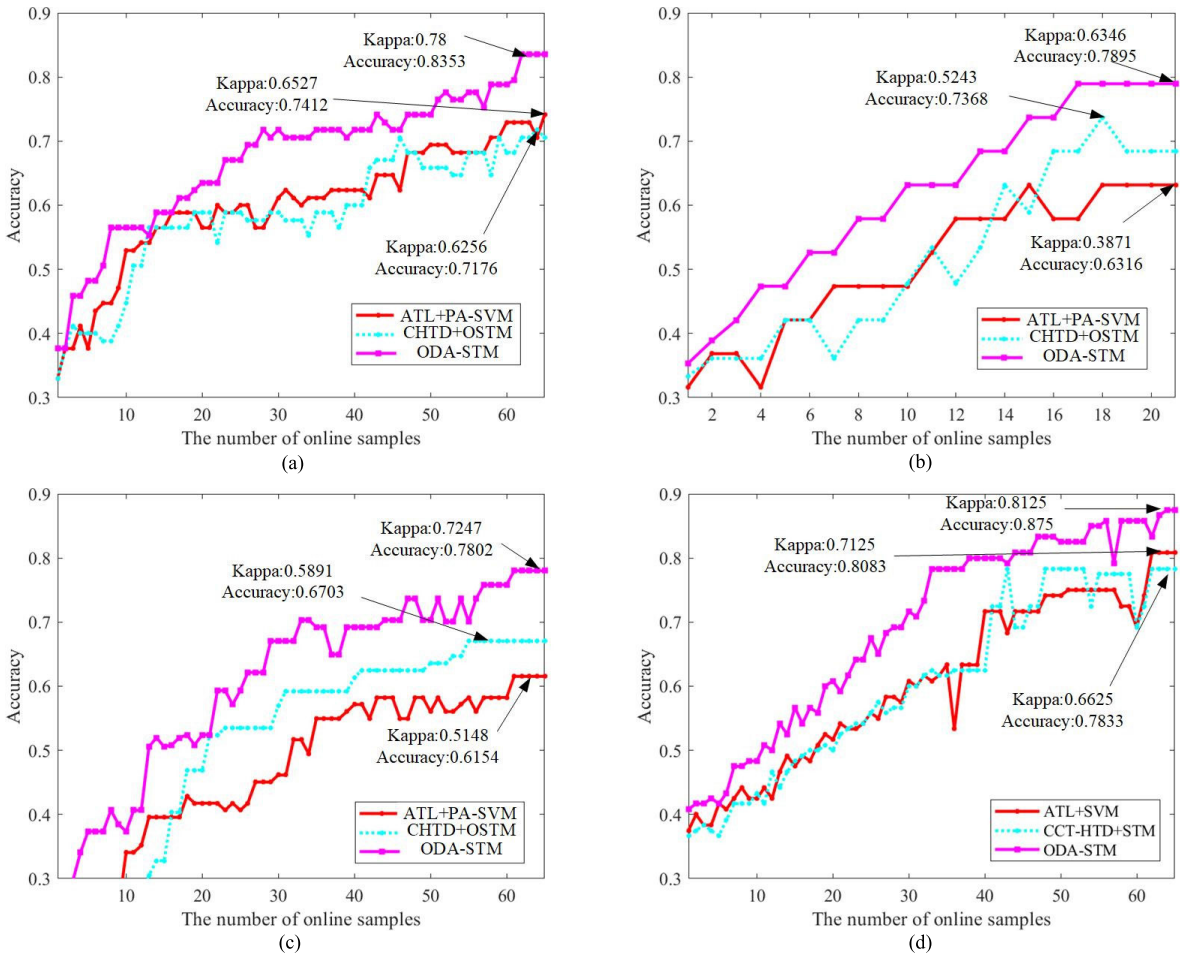


Fig. 11. Accuracy of ODA-STM with different numbers of online samples. (a)–(d) display the accuracy of different online classifiers using datasets 1, 2, 3, and 4, respectively.

E. Evaluation of the Performance of the ODA-STM Compared With Typical Online Classifiers

To further evaluate the effectiveness of the online updating mechanism of ODA-STM, the experiments are conducted on four datasets to examine the ODA-STM performance as the acquisition of sequential cross-source samples. For the competitors, since the ATL and CHTD have obtained better results for cross-source recognition, these two methods are selected to combine the online version of SVM (i.e., PA-SVM [43]) and the online version of STM (i.e., OSTM [42]) as the comparison methods, respectively. In the experiments, since PA-SVM and OLSTM cannot handle cross-source data directly, ATL and CHTD are applied to extract domain adaptation features. The dataset is split into equal training and test sets. First, all the methods adopt the random model parameters. Then, the accuracies of PA-SVM, OLSTM, and the proposed ODA-STM are examined as the updating of sequential randomly selected training samples, and the results are displayed in Fig. 11.

As seen from Fig. 11, since the model parameter is initialized randomly, these methods get low recognition accuracies at first. With the update of online samples, the recognition accuracies of these methods are improved. For datasets 1, 3, and 4, the accuracy of ODA-STM is steadily better than 70% when the number of sequence samples exceeds 50. Then, with the acquisition of sequence samples, the recognition accuracy achieves results better than 80%. For dataset 2, the results of different methods are significantly improved for a small number of sequence samples, which is due to the fact that dataset 2 has a smaller number of categories, making it easier to learn an effective classification hyperplane. It is noted that the proposed ODA-STM can achieve the highest recognition accuracy in most cases with the same number of sequence samples. Moreover, the best recognition results of these methods are marked in Fig. 11. It is seen that our ODA-STM obtains better recognition accuracy and Kappa coefficient than competitors. This is due to the fact that ODA-STM can dynamically adjust the parameters of cross-source subspace and hyperplane simultaneously, while the comparison method relies on fixed transferred features and lacks flexibility. This finding indicates that ODA-STM can efficiently utilize sequence samples to gradually improve object recognition performance, which is suitable for multi-source satellite sequence observations.

F. Evaluation of the Performance of the DA-STM Compared With Typical Deep Learning Methods

As the representatives, deep learning methods have been widely applied in cross-source recognition. To further evaluate the performance of DA-STM compared with state-of-the-art deep learning methods, experiments are conducted on dataset 1

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TABLE VIII
OA OF DEEP LEARNING METHODS AND THE DA-STM ACROSS SOURCES FOR DATASET 1

	P=0.2	P=0.3	P=0.4	P=0.5
DANN	48.3%	53.6%	60.7%	71.7%
DIDA	41.7%	59.2%	62.9%	72.4%
CDAC	28.0%	30.4%	52.5%	68.5%
UDAGCN	47.1%	57.1%	65.5%	69.3%
DA-STM	51.77%	77.12%	79%	81.18%

TABLE IX
OA OF DEEP LEARNING METHODS AND THE DA-STM ACROSS SOURCES FOR DATASET 2

	P=0.2	P=0.3	P=0.4	P=0.5
DANN	47.2%	52.8%	57.4%	70.6%
DIDA	42.9%	58.3%	66.7%	69.5%
CDAC	33.3%	50.0%	63.9%	80.6%
UDAGCN	42.9%	57.4%	61.9%	68.7%
DA-STM	58.62%	69.23%	77.27%	83.33%

and dataset 2 to compare our DA-STM with three domain adaption deep learning methods, i.e., DIDA, DANN, CDAC, and UDAGCN [56] networks. In the experiments, different proportions of samples from the dataset are applied to be used as training samples, and the rest samples are used as test set. For the deep learning methods, the learning rate is set to 5×10^{-4}, and the batch size is set to 4. The obtained results are displayed in Tables VIII and IX.

Where p denotes the proportion of training set for the whole dataset. From Tables VIII and IX, it can be seen that these methods get different results under different ratios of training samples. Although these deep learning methods have excellent feature representation and discriminative power, it relays a large number of samples. When the training sample set is small, the deep learning methods obtain poor results (see Tables VIII and IX). Note that the proposed DA-STM has much smaller parameters than deep learning methods and is thus suitable for the small-sample case. Moreover, the DA-STM gets obviously better results as the training samples increase. It indicates the effectiveness of the DA-STM, especially for the small-sample case.

VI. CONCLUSION

To recognize objects described by cross-source remote sensing data effectively, the DA-STM method is proposed to generate distinct cross-source classification hyperplane by adapting to the heterogeneous structures and diverse distribution of cross-source data simultaneously, thus supporting an effective cross-source object recognition. Compared with the typical STM methods that rely on single-source homogeneous projection tensor, the proposed DA-STM can exploit the cross-source heterogeneous projection tensors constrained by shared core tensor to excavate the shared information of multisource heterogeneous data. Unlike the existing STM can only learn hyperplane from labeled samples, the proposed DA-STM can absorb useful information from both labeled and unlabeled samples via adaptive class labels. In addition, to obtain the efficient training of DA-STM, the DA-STM-oriented decomposition method is proposed to break the original complex quadratic programming into a series of analytical optimizations. To improve the classifier performance gradually with the acquisition of sequence sample, the online extension of DA-STM (i.e., ODA-STM) is presented to ensure updating parameters of the model online. Using cross-angle, cross-resolution, and cross-scene remote sensing images, experimental results demonstrate that the proposed DA-STM outperforms the typical domain adaption SVM and STM methods. Not only that, the DA-STM can get better accuracies than typical deep learning methods under small-sample cases.

To further enhance the superiority of the DA-STM, future works will focus on extending the current DA-STM to a nonlinear version to achieve domain adaption in a higher-dimensional space and constructing the outlier control scheme to adjust the weight of samples automatically.

APPENDIX

A. Updating of $\boldsymbol{W}_2^l$ for DA-STM

When optimizing the $\boldsymbol{W}_2^l$, the remained variables are fixed. The current sub-optimization is:

$$\begin{aligned}&\min_{\boldsymbol{W}_2^l, b_2} \frac{1}{2}\left\|\mathcal{G}\prod_{l=1}^{L}\times_l \boldsymbol{W}_2^l\right\|_F^2 + C_2\sum_{i=1}^{N_2}\xi_2^i + C_3\sum_{i=N_2+1}^{N_2+N_{2_ul}}\xi_2^i\\ &\text{s.t. } y_2^i\left(\left\langle \mathcal{G}\prod_{l=1}^{L}\times_l \boldsymbol{W}_2^l, \mathcal{X}_2^i\right\rangle + b_2\right) \geq 1-\xi_2^i\\ &\quad \xi_2^i \geq 0, \quad 1 \leq i \leq N_2 + N_{2_ul}. \end{aligned} \tag{42}$$

Then, let $\widetilde{\boldsymbol{W}}_2^l = \boldsymbol{W}_2^l \boldsymbol{K}^{(1/2)}$, where

$$\boldsymbol{K} = \mathrm{Mat}_{(l)}\left(\mathcal{G}\prod_{j\neq l}\times_j \boldsymbol{W}_2^j\right)\mathrm{Mat}_{(l)}^T\left(\mathcal{G}\prod_{j\neq l}\times_j \boldsymbol{W}_2^j\right). \tag{43}$$

Let $\widetilde{\boldsymbol{X}_i} = \boldsymbol{K}^{-(1/2)} \times \mathrm{Mat}_{(l)}(\mathcal{G}\prod_{j\neq l}\times_j \boldsymbol{W}_2^j)\mathrm{Mat}_{(l)}^T(\mathcal{X}_2^i)$, $1 \leq i \leq N_2 + N_{2_ul}$. The intervariable can be rewritten by vector form: $\widetilde{\boldsymbol{w}}_2^l = \mathrm{vec}(\widetilde{\boldsymbol{W}}_2^l)$, $\widetilde{\boldsymbol{x}_i} = \mathrm{vec}(\widetilde{\boldsymbol{X}_i})$.

Take the above equation into (42). We have:

$$\begin{aligned}&\min_{\widetilde{\boldsymbol{w}}_2^l, b_2} \frac{1}{2}\widetilde{\boldsymbol{w}}_2^{lT}\widetilde{\boldsymbol{w}}_2^l + C_2\sum_{i=1}^{N_2}\xi_2^i + C_3\sum_{i=N_2+1}^{N_2+N_{2_ul}}\xi_2^i\\ &\text{s.t. } y_2^i\left(\widetilde{\boldsymbol{w}}_2^{lT}\widetilde{\boldsymbol{x}}_i + b_2\right) \geq 1-\xi_2^i\\ &\quad \xi_2^i \geq 0, \quad 1 \leq i \leq N_2 + N_{2_ul}. \end{aligned} \tag{44}$$

To remove the constraints of above equation, the Lagrangian function can be listed as follows:

$$\begin{aligned}&\mathcal{L}\left(\widetilde{\boldsymbol{w}}_2^l, b_2, \xi_2^i, \alpha_i, \lambda_i\right)\\ &= \frac{1}{2}\widetilde{\boldsymbol{w}}_2^{lT}\widetilde{\boldsymbol{w}}_2^l + C_2\sum_{i=1}^{N_2}\xi_2^i\\ &\quad + C_3\sum_{i=N_2+1}^{N_2+N_{2_ul}}\xi_2^i - \sum_{i=1}^{N_2+N_{2_ul}}\alpha_i\left(y_2^i\left(\widetilde{\boldsymbol{w}}_2^{lT}\widetilde{\boldsymbol{x}}_i + b_2\right) - 1 + \xi_2^i\right)\\ &\quad - \sum_{i=1}^{N_2+N_{2_ul}}\lambda_i\xi_2^i \end{aligned} \tag{45}$$

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where α_i, λ_i denote Lagrangian multipliers. Let partial derviatives of Eq. (45) respect to $\boldsymbol{W}_1^l, b, \xi_1^i$ be zeros. We have:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \widetilde{\boldsymbol{w}}_2^l} = 0 &\Rightarrow \widetilde{\boldsymbol{w}}_2^l = \sum_{i=1}^{N_2+N_{2_ul}} \alpha_i y_2^i \widetilde{\boldsymbol{x}}_i \\
\frac{\partial \mathcal{L}}{\partial b_2} = 0 &\Rightarrow \sum_{i=1}^{N_2+N_{2_ul}} y_2^i \alpha_i = 0 \\
\frac{\partial \mathcal{L}}{\partial \xi_1^i} = 0 &\Rightarrow \alpha_i + \lambda_i = C_2, \quad 1 \le i \le N_2 \\
\alpha_i + \lambda_i &= C_3, \quad N_2 + 1 \le i \le N_2 + N_{2_ul}.
\end{aligned} \tag{46}$$

By applying the above equations, we can get dual problem as

$$\begin{aligned}
&\min_{\boldsymbol{\alpha}} \boldsymbol{\alpha}^T \boldsymbol{K} \boldsymbol{\alpha} - \boldsymbol{e}^T \boldsymbol{\alpha} \\
&\text{s.t.} \sum_{i=1}^{N_2+N_{2_ul}} y_2^i \alpha_i = 0 \\
&\quad 0 \le \alpha_i \le C_2, \quad 1 \le i \le N_2 \\
&\quad 0 \le \alpha_{i+N_2} \le C_3, \quad 1 \le i \le N_{2_ul}
\end{aligned} \tag{47}$$

where $\boldsymbol{K}(i,k) = (1/2) y_2^i y_2^k \widetilde{\boldsymbol{x}}_i^T \widetilde{\boldsymbol{x}}_k$. Utilize the interior point algorithm [38] or the decomposition algorithm presented in Section III-C to solve the above optimization. Then, according to $\widetilde{\boldsymbol{w}}_2^l = \sum_{i=1}^{N_2+N_{2_ul}} \alpha_i y_2^i \widetilde{\boldsymbol{x}}_i$ and $\widetilde{\boldsymbol{W}}_2^l = \boldsymbol{W}_2^l \boldsymbol{K}^{(1/2)}$, the factor matrix $\boldsymbol{W}_2^l$ for source 2 data can be updated.

B. Parameter Updating of ODA-STM

For ODA-STM, assume that the sample obtained in tth round is from source 1. The alternating updating strategy is adopted for the ODA-STM to update $\boldsymbol{W}_1^l$, $\mathcal{G}$, and b_1 sequentially.

When optimizing the factor matrix $\boldsymbol{W}_1^l$, introduce the intermediator variables:

$$\begin{aligned}
\widetilde{\boldsymbol{W}}_1^l &= \boldsymbol{W}_1^l \boldsymbol{K}^{\frac{1}{2}} \\
\widetilde{\boldsymbol{A}} &= \boldsymbol{K}^{-\frac{1}{2}} \times \mathrm{Mat}_{(l)} \left(\mathcal{G} \prod_{j \ne l} \times_j \boldsymbol{W}_1^j \right) \mathrm{Mat}_{(l)}^T \left(\widehat{\mathcal{G}} \prod_{l=1}^{L} \times_l \widehat{\boldsymbol{W}}_1^l \right) \\
\widetilde{\boldsymbol{X}_t} &= \boldsymbol{K}^{-\frac{1}{2}} \times \mathrm{Mat}_{(l)} \left(\mathcal{G} \prod_{j \ne l} \times_j \boldsymbol{W}_1^j \right) \mathrm{Mat}_{(l)}^T \left(\mathcal{X}_1^t \right)
\end{aligned} \tag{48}$$

where $\boldsymbol{K}$ is defined by $\boldsymbol{K} = \mathrm{Mat}_{(l)}(\mathcal{G} \prod_{j \ne l} \times_j \boldsymbol{W}_1^j) \mathrm{Mat}_{(l)}^T (\mathcal{G} \prod_{j \ne l} \times_j \boldsymbol{W}_1^j)$.

The vector form of the intermediator variables can be expressed as $\widetilde{\boldsymbol{w}}_1^l = \mathrm{vec}(\widetilde{\boldsymbol{W}}_1^l)$, $\widetilde{\boldsymbol{a}} = \mathrm{vec}(\widetilde{\boldsymbol{A}})$, and $\widetilde{\boldsymbol{x}}_t = \mathrm{vec}(\widetilde{\boldsymbol{X}_t})$. Then, the current sub-optimization is derived as

$$\begin{aligned}
&\min_{\widetilde{\boldsymbol{w}}_1^l, b_1} \frac{1}{2} \widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{w}}_1^l - \widetilde{\boldsymbol{w}}_1^l \widetilde{\boldsymbol{a}} + \frac{1}{2} \left\| b_1 - \widehat{b}_1 \right\|_F^2 + C_1 \times \xi_1 \\
&\text{s.t. } y_1^i \left(\widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{x}}_t + b_1 \right) \ge 1 - \xi_1, \quad \xi_1 \ge 0.
\end{aligned} \tag{49}$$

The corresponding Lagrangian function can be obtained as

$$\begin{aligned}
&\mathcal{L}\left(\widetilde{\boldsymbol{w}}_1^l, b_1, \xi_1, \alpha, \lambda \right) \\
&= \frac{1}{2} \widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{w}}_1^l - \widetilde{\boldsymbol{w}}_1^l \widetilde{\boldsymbol{a}} + \frac{1}{2} \left\| b_1 - \widehat{b}_1 \right\|_F^2 \\
&\quad + C_1 \times \xi_1 - \alpha \left(y_1^t \left(\widetilde{\boldsymbol{w}}_1^{lT} \widetilde{\boldsymbol{x}}_t + b_1 \right) - 1 + \xi_1 \right) - \lambda \xi_1
\end{aligned} \tag{50}$$

where α and λ denote the dual variables. Let the partial derivatives of the Lagrangian function with respect to the primal variables be zeros. We have:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \widetilde{\boldsymbol{w}}_1^l} = 0 &\Rightarrow \widetilde{\boldsymbol{w}}_1^l = \alpha y_1^t \widetilde{\boldsymbol{x}}_t + \widetilde{\boldsymbol{a}} \\
\frac{\partial \mathcal{L}}{\partial b_1} = 0 &\Rightarrow b_1 = \widehat{b}_1 + \alpha y_1^t \\
\frac{\partial \mathcal{L}}{\partial \xi_1} = 0 &\Rightarrow C_1 - \alpha - \lambda = 0.
\end{aligned} \tag{51}$$

Substituting the above equations into (50), the obtained dual problem is shown below

$$\begin{aligned}
&\min_{\alpha} \frac{(\alpha)^2}{2} \widetilde{\boldsymbol{x}}_t^T \widetilde{\boldsymbol{x}}_t + \alpha y_1^t \widetilde{\boldsymbol{x}}_t^T \widetilde{\boldsymbol{a}} + \frac{(\alpha)^2}{2} + \alpha y_1^t \widehat{b}_1 - \alpha \\
&\text{s.t. } 0 \le \alpha \le C_1.
\end{aligned} \tag{52}$$

Let the derivative of the objective function of the above equation with respect to α be zero. We have:

$$\alpha^* = \frac{1 - y_1^t \widehat{b}_1 - y_1^t \widetilde{\boldsymbol{x}}_t^T \widetilde{\boldsymbol{a}}}{\widetilde{\boldsymbol{x}}_t^T \widetilde{\boldsymbol{x}}_t + 1}. \tag{53}$$

Combining with the boundary constraint, the closest feasible solution to α^* is chosen as the optimized α, that is:

$$\alpha = \max\left(0, \min\left(C_1, \alpha^*\right)\right). \tag{54}$$

The results of the (54) is optimal α, which can be used to calculate the updated factor matrix and bias b_1 by

$$\begin{aligned}
\widetilde{\boldsymbol{w}}_1^l &= \alpha y_1^t \widetilde{\boldsymbol{x}}_t + \widetilde{\boldsymbol{a}} \\
\boldsymbol{W}_1^l &= \widetilde{\boldsymbol{W}}_1^l \boldsymbol{K}^{-\frac{1}{2}} \\
b_1 &= \widehat{b}_1 + \alpha y_1^t.
\end{aligned} \tag{55}$$

When optimizing the shared core tensor, the following variable substitutions are adopted:

$$\begin{aligned}
\widetilde{\boldsymbol{g}} &= \mathrm{vec}(\mathcal{G}) \boldsymbol{K}^{\frac{1}{2}} \\
\boldsymbol{K} &= \left(\boldsymbol{W}_1^1 \otimes \cdots \otimes \boldsymbol{W}_1^L \right) \left(\boldsymbol{W}_1^1 \otimes \cdots \otimes \boldsymbol{W}_1^L \right)^T \\
&\quad + \left(\widehat{\boldsymbol{W}}_2^1 \otimes \cdots \otimes \widehat{\boldsymbol{W}}_2^L \right) \left(\widehat{\boldsymbol{W}}_2^1 \otimes \cdots \otimes \widehat{\boldsymbol{W}}_2^L \right)^T \\
\widehat{\boldsymbol{K}} &= \left(\boldsymbol{W}_1^1 \otimes \cdots \otimes \boldsymbol{W}_1^L \right) \left(\widehat{\boldsymbol{W}}_1^1 \otimes \cdots \otimes \widehat{\boldsymbol{W}}_1^L \right)^T \\
&\quad + \left(\widehat{\boldsymbol{W}}_2^1 \otimes \cdots \otimes \widehat{\boldsymbol{W}}_2^L \right) \left(\widehat{\boldsymbol{W}}_2^1 \otimes \cdots \otimes \widehat{\boldsymbol{W}}_2^L \right)^T
\end{aligned} \tag{56}$$

$$\widetilde{\boldsymbol{a}} = \boldsymbol{K}^{-\frac{1}{2}} \widehat{\boldsymbol{K}} \mathrm{vec}\left(\widehat{\mathcal{G}} \right), \widetilde{\boldsymbol{x}}_t = \boldsymbol{K}^{-\frac{1}{2}} \mathrm{vec}\left(\mathcal{X}_1^t \prod_{l=1}^{L} \times_l \boldsymbol{W}_1^l \right) \tag{57}$$

where the operator $\otimes$ denotes the Kronecker product. Then, the optimization problem can be simplified as

$$\begin{aligned}
&\min_{\widetilde{\boldsymbol{g}}, b_1} \frac{1}{2} \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{g}} - \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{a}} + \frac{1}{2} \left\| b_1 - \widehat{b}_1 \right\|_F^2 + C_1 \times \xi_1 \\
&\text{s.t. } y_1^t \left(\widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{x}}_t + b_1 \right) \ge 1 - \xi_1 \\
&\quad \xi_1 \ge 0.
\end{aligned} \tag{58}$$

The corresponding Lagrangian function is shown below:

$$\mathcal{L}\left(\widetilde{\boldsymbol{g}}, b_1, \xi_1, \alpha, \lambda \right) = \frac{1}{2} \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{g}} - \widetilde{\boldsymbol{g}}^T \widetilde{\boldsymbol{a}} + \frac{1}{2} \left\| b_1 - \widehat{b}_1 \right\|_F^2 + C_1 \xi_1$$

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$$-\alpha\left(y_1^t\left(\widetilde{\boldsymbol{g}}^T\widetilde{\boldsymbol{x}}_t+b_1\right)-1+\xi_1\right)-\lambda\xi_1. \quad (59)$$

Let the partial derivative of the above equation with respect to the primal variables be zeros. We have:

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\widetilde{\boldsymbol{g}}}&=0\Rightarrow\widetilde{\boldsymbol{g}}=\widetilde{\boldsymbol{a}}+\alpha y_1^t\widetilde{\boldsymbol{x}}_t\\ \frac{\partial\mathcal{L}}{\partial b_1}&=0\Rightarrow b_1=\widehat{b}_1+\alpha y_1^t\\ \frac{\partial\mathcal{L}}{\partial\xi_1}&=0\Rightarrow C_1-\alpha-\lambda=0.\end{aligned} \quad (60)$$

Substitute these equations into (59). We can list the dual problem as

$$\begin{aligned}&\min_{\alpha}\frac{(\alpha)^2}{2}\widetilde{\boldsymbol{x}}_t^T\widetilde{\boldsymbol{x}}_t+\alpha y_1^t\widetilde{\boldsymbol{x}}_t^T\widetilde{\boldsymbol{a}}+\frac{(\alpha)^2}{2}+\alpha y_1^t\widehat{b}_1-\alpha\\ &\text{s.t. } 0\le\alpha\le C_1.\end{aligned} \quad (61)$$

Analogous to the updating of the factor matrix, we give the optimal solution of (61) directly as follows:

$$\alpha=\max\left(0,\min\left(C_1,\frac{1-y_1^t\widehat{b}_1-y_1^t\widetilde{\boldsymbol{x}}_t^T\widetilde{\boldsymbol{a}}}{\widetilde{\boldsymbol{x}}_t^T\widetilde{\boldsymbol{x}}_t+1}\right)\right). \quad (62)$$

Then, use (60) to achieve the updating of core tensor $\mathcal{G}$ and bias b_1. Repeatedly update the factor matrix, core tensor, and the bias until the number of iterations exceeds a specific preset constant or the iteration difference is less than a specific threshold. Then, output $\boldsymbol{W}_1^l$, $\boldsymbol{W}_2^l$, $\mathcal{G}$, b_1, and b_2 as the final updated model parameters of ODA-STM.

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